

High-dimensional state trajectories reveal transformer inference dynamics

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Abstract

Mechanistic studies localize language-model function in parameters, heads or neurons, but those parts do not specify how one inference unfolds. We instead treat the ordered layer-state trajectory as the empirical process. Across Qwen, Llama and Gemma, realized order was 1.60–2.24 times more chord-aligned than endpoint-preserving shuffles. Random initializations were still more aligned, whereas reordered executions retained geometric regularity but lost native outputs: architecture supplies a propagation scaffold, training organizes compatible task coordinates and native order executes functionally consequential transitions. Across Pythia training, fixed task coordinates emerged only after training began and function required compatible input, block and output components. Ordered-prefix summaries did not generally predict final boundaries beyond current state. After additive interventions failed, confidence-gated local-transition actuators crossed a declared full-vocabulary boundary in 17/87, 13/87 and 9/87 held-out prompts, with 0/203, 2/203 and 0/203 non-target changes. Ordered high-dimensional states expose an observable and controllable computational process, while geometry, forecasting, action assignment and correctness remain bounded.

Keywords: transformer inference; mechanistic interpretability; layer-state trajectories; chord-directed order; structured residual motion; internal dynamics

Introduction

Large language models transform their internal state many times before producing a token. Existing work has localized components, decoded intermediate outputs and identified task-specific structures^{5-12,18,19}, but these observations have not yet converged on a common dynamical description of inference. Model scale is one reason. Another may be more basic: an explanation depends on what is measured and on the coordinates used to compare those measurements.

Interpretability methods expose different aspects of transformer computation. Probes test decodable information, while circuit analyses isolate local computations. Causal tracing and model editing localize factual associations. Other methods use vocabulary readbacks or sparse activation features⁵⁻¹². Their conclusions depend on probe capacity, layer selection and the chosen projection. Dimensionality and similarity metrics also affect representation comparisons^{6,14-17}. It remains uncertain which object can be compared across layers and checkpoints while preserving the process being measured.

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Intermediate vocabulary methods address several distinct questions. The tuned lens calibrates layerwise token predictions, whereas DoLa contrasts logits from different layers to improve decoding^{11,20}. We use the same broad access route for a different purpose. Our question is whether vocabulary-facing coordinates preserve mechanism-relevant geometry because of their learned row identities. Row permutation and matched random projection make that claim experimentally separable from the usefulness of token-level readout.

We examined whether part of this difficulty is an object-coordinate mismatch. Earlier work has decoded intermediate predictions, compared representation geometry and traced particular computations^{11,15–20}. We treated the choice of empirical object as part of the mechanism claim and tested it directly. The decisive question was whether a proposed coordinate retained its explanatory role when its learned identity was removed. This turns coordinate choice from an analytical convention into a falsifiable hypothesis.

The experimental main line was developed in Qwen2.5-1.5B-Instruct. Gemma-2-2B-it and Llama-3.2-1B-Instruct were then used for cross-checkpoint replication and boundary tests. The three checkpoints differ substantially in depth, hidden width and vocabulary size; cross-model panels should therefore be read as tests of recurrence under architectural variation, not as three independent discovery sequences.

Study checkpoints

Main line: Qwen/Qwen2.5-1.5B-Instruct
1.5B | 28 layers | d=1536 | vocab=151,936 | context=32,768*
Cross-model: google/gemma-2-2b-it
2B | 26 layers | d=2304 | vocab=256,000 | context=8,192
Cross-model: meta-llama/Llama-3.2-1B-Instruct
1B | 16 layers | d=2048 | vocab=128,256 | context=131,072
*Qwen tokenizer model_max_length=131,072; model_max_position_embeddings=32,768.

The study began with TopK vocabulary readbacks because they made intermediate states inspectable at every layer. Their changes across depth exposed a reproducible late transition and raised questions about recurrent transformations and functional windows. We then tested the apparent privilege of the readback. If learned vocabulary identity supplied the relevant geometry, changing row identity should destroy it. Row permutation left preservation nearly unchanged, while matched random projections often preserved as much or more. This result rejected the learned-row explanation and shifted the empirical object to the hidden states from which the readbacks were derived. Grouped, leakage-controlled and capacity-matched comparisons established ordered high-dimensional layer states as stronger coordinates for controlled mechanism-label organization.

We then asked which parts of this organization came from architecture, training and realized execution. Real layer order was more chord-directed than endpoint-preserving shuffles, but random initializations were still more aligned. Reversed and fixed-permutation executions retained geometric regularity while losing native outputs, and a radial audit showed that the metric was scale-sensitive. Pythia component swaps and training checkpoints then separated architecture-compatible propagation from training-organized task coordinates and component compatibility. Finally, mechanism-conditioned local-transition actuators moved held-out trajectories and crossed a declared emitted-token boundary. The resulting account is process-level and causal within tested boundaries, not a universal dynamical law.

The study makes four testable contributions. It establishes the ordered layer-state trajectory as a common empirical process object; distinguishes architecture-supported propagation from training-organized task coordinates and native ordered execution; bounds predictive claims by showing that ordered history does not generally add information beyond current state; and demonstrates selective trajectory and emitted-token

boundary control with an abstaining actuator. Each contribution is tied to a projection, order, metric, training or intervention control.

Core concepts

The mechanism picture uses a small set of coordinates. Their roles are fixed throughout the paper.

Term	Operational meaning
Layer-state trajectory	The ordered sequence of hidden states across transformer layers for one checkpoint and prompt. “Hidden-state trajectory” denotes the same object; “layer-state trajectory” is the canonical term used in claims.
Functional window	A checkpoint-specific layer interval selected by a declared functional objective rather than by nominal layer identity.
Candidate trajectories	The leading and competing candidate-relative paths declared by the controlled prompt construction.
ΔU	A signed low-rank scalar measuring relative support for the leading versus competing candidate trajectory within the decision window.
Observed-transition coordinate O_l^{cont}	A continuous latent coordinate derived from the realized transition signature from layer l to $l + 1$; it describes rather than predicts that transition.
Target-excluded estimate $\hat{O}_l^{\text{cont}}$	An estimate of the latent transition coordinate constructed without target-layer information; it tests advance recoverability.
Chord-directed order	Greater endpoint-chord alignment and lower detour or turning curvature than the declared order-shuffle control under a specified metric; not an exact geodesic or a learned-function signature.
Structured residual	Off-reference-direction motion that retains stable candidate or mechanism information beyond the aligned component.

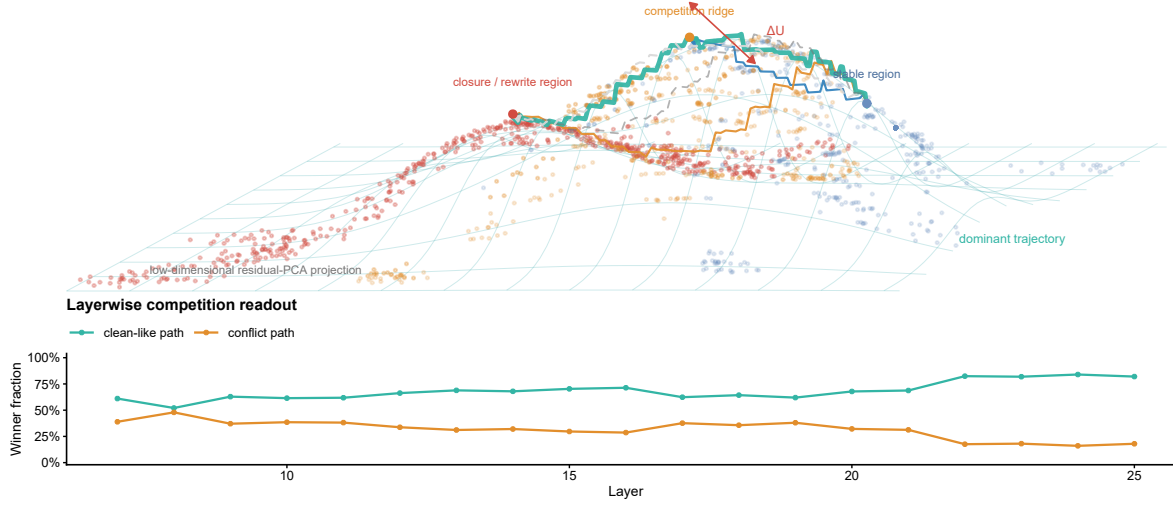
Results overview

This study separates three levels of internal transformer organization. Architecture supports relational propagation, training organizes compatible task coordinates across input, transition and output components, and native layer order executes functionally consequential transitions. Chord-directed geometry recurs across tasks and checkpoints but is stronger at random initialization, survives function-destroying reorderings and is sensitive to radial scale; it is therefore a bounded metric observation rather than a learned-reasoning signature. Structured residual motion retains complementary information, and mechanism-conditioned local-transition actuators produce selective held-out changes. The ordered hidden-state trajectory provides the common empirical object on which these levels can be compared and perturbed.

The graphical overview gives a low-dimensional orientation aid. The surface, paths and competition ridge are illustrative; the evidence comes from high-dimensional layer-state arrays, projection controls, grouped comparisons, route-order nulls and held-out operational tests. The Results then follow the discovery sequence. Vocabulary readbacks first supplied an observational window, operator and depth controls narrowed that interpretation, and projection controls rejected learned vocabulary-row identity as the source of the preserved geometry. That failed explanation exposed the ordered hidden-state trajectory as the more fundamental empirical object.

Low-dimensional orientation of candidate-path competition

low-dimensional orientation aid; not the primary mechanism representation



Graphical overview | Candidate-trajectory competition during transformer inference. This low-dimensional residual-PCA display is an orientation aid for the retained conclusion. It illustrates layerwise competition, the bounded dominance observable ΔU , and the distinction between dominant and competing paths. The surface and paths are not literal estimates of a semantic manifold, and the mechanism claim does not rest on this projection. Subsequent figures test the account in declared high-dimensional layer-state coordinates and under projection, order-shuffle, leakage and intervention controls.

Quick guide to the evidence chain

Figures	Question answered	Retained result
Figs. 1-4	What can vocabulary-facing readbacks reveal across layers and checkpoints?	A reproducible observational flow and coarse late transition motivate functional alignment, while fine operator labels largely track depth.
Figs. 5-7	Can candidate competition be compressed and detected before the decision window?	ΔU captures candidate-relative dominance, with leakage-reduced precursor organization that varies by checkpoint.
Fig. 8	Does learned vocabulary-row identity supply the observed geometry?	Row permutation and matched projection controls reject learned row identity as the source.
Figs. 9-11	Which empirical object best retains the mechanism organization?	Grouped and matched-capacity audits upgrade the primary object to the ordered high-dimensional layer-state trajectory.
Figs. 12-14	What dynamical organization does the trajectory exhibit?	Descriptive transition closure, geodesic-biased flow and structured non-geodesic residual motion define the retained mechanism picture.
Fig. 15	Is the trajectory organization operationally discriminating?	Trajectory-informed policies produce target-specific internal modulation beyond fixed and matched shuffled alternatives.

1 Opening a layer-resolved observational window

Could vocabulary-facing readbacks make hidden computation observable?

The first obstacle was observational. Intermediate activations were available, but there was no agreed coordinate for inspecting their evolution without first choosing a circuit, label or task-specific probe. The vocabulary projection attached to each checkpoint offered a possible window. If a TopK readback retained organized information, its layerwise profile should recur across seeds and remain recognizable under

changes to TopK and prompt form. Projecting each hidden state through the output projection produced a layer-indexed vocabulary neighbourhood that made this premise testable (Fig. 1a). It was our first candidate coordinate, not yet a causal object.

Vocabulary readbacks opened a layer-resolved observational window

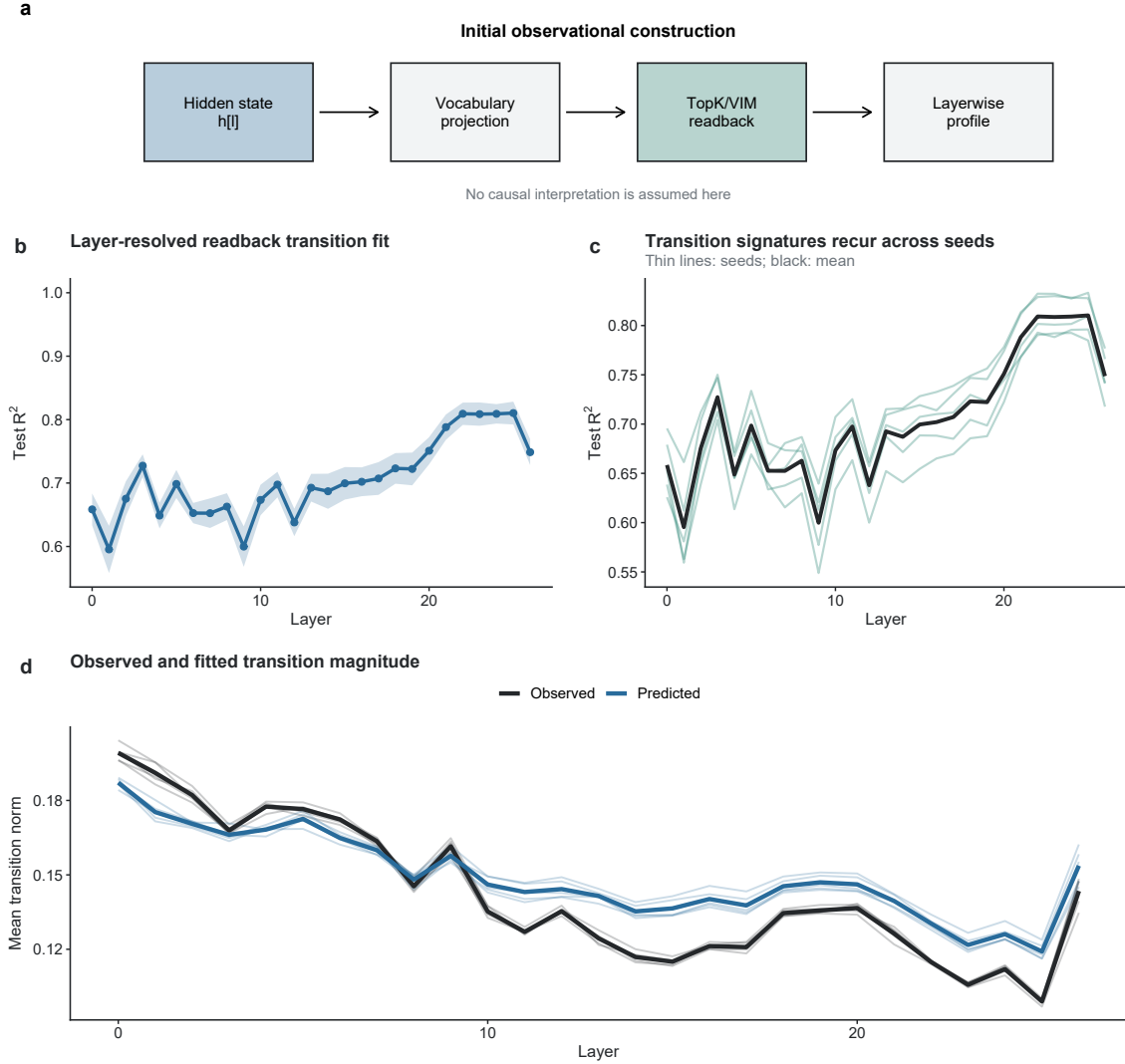


Fig. 1: Vocabulary readbacks as a layer-resolved observational window. **a**, Construction schematic showing the mapping from a hidden state at layer l to a vocabulary projection, TopK/VIM readback and layerwise profile. The schematic defines an observable and does not assign causal status to the projection. **b**, Mean held-out transition-fit R^2 by layer; ribbons show 95% confidence intervals across five seeds. Each split contained 196 observations, with 147 training and 49 test observations. **c**, Layerwise transition signatures for individual seeds (thin teal lines) and their mean (black line). **d**, Observed and fitted mean transition magnitudes across layers; thin lines show individual seeds and thick lines show seed means. See the corresponding Source Data figure and the Source Data manifest.

The vocabulary-facing neighbourhoods changed reproducibly with layer depth. Their profiles recurred across the available seeds and survived the tested TopK and prompt perturbations (Figs. 1 and 2). Natural-language transformations preserved high centre cosine while producing larger variation in overlap and path agreement, separating stable readback geometry from condition-sensitive neighbourhood composition (Fig. 2d). Complete prompt, seed, TopK and negative-example summaries are reported in Supplementary Figs. S1-S3.

Readback-flow organization survived available robustness checks

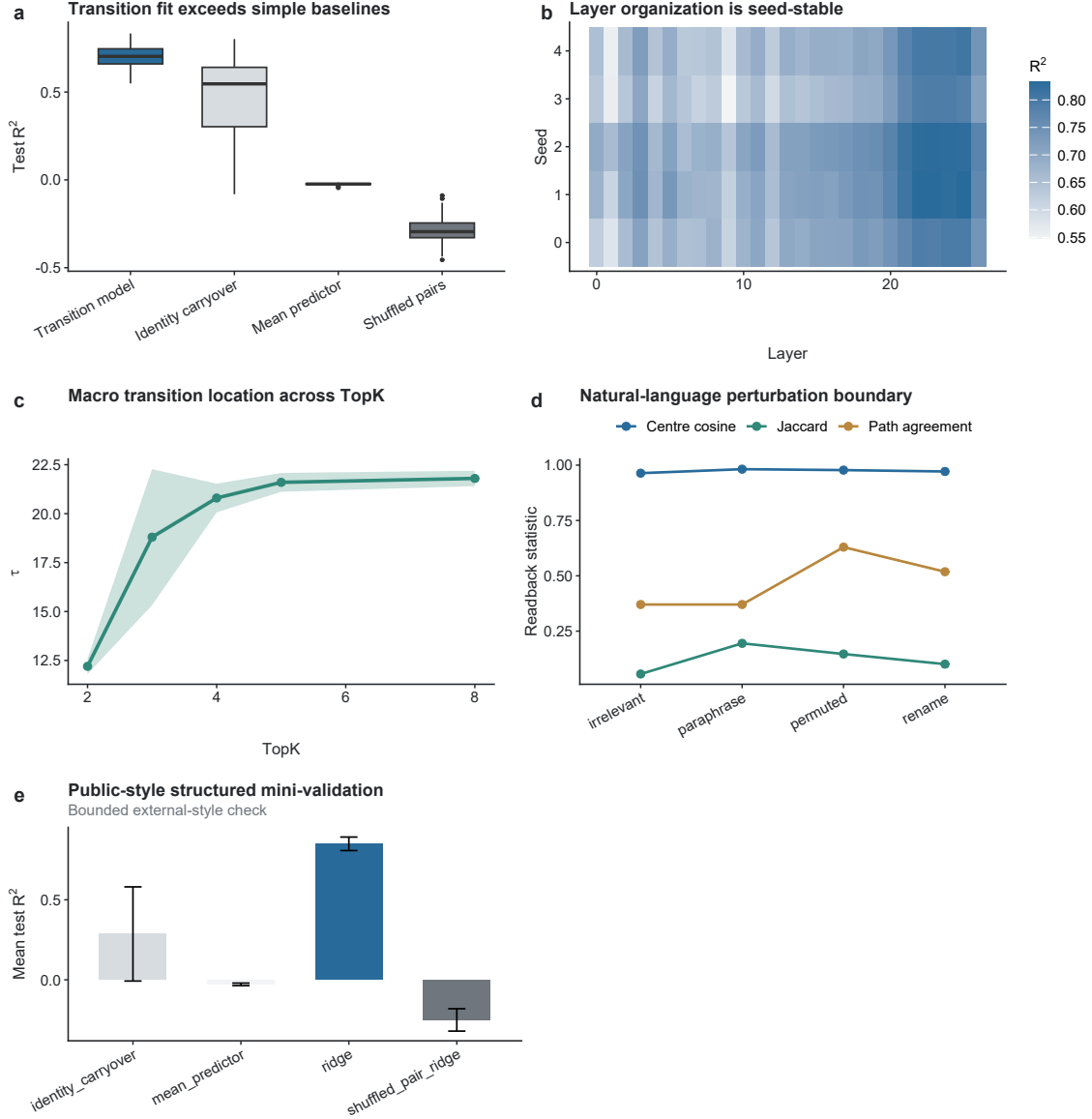
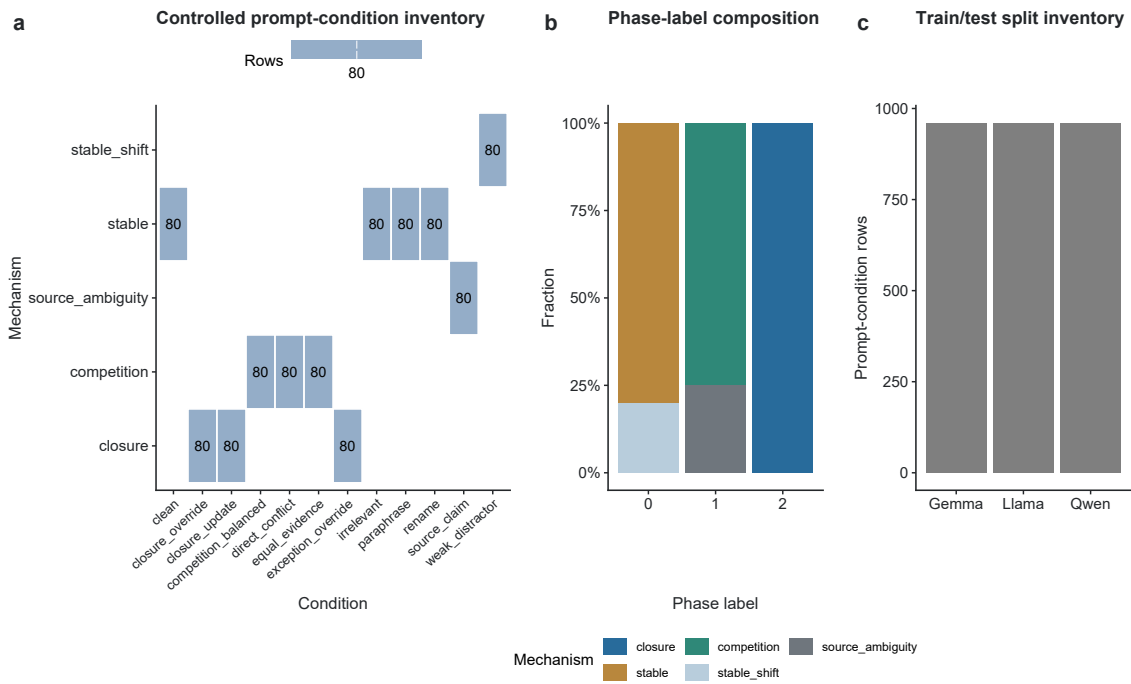


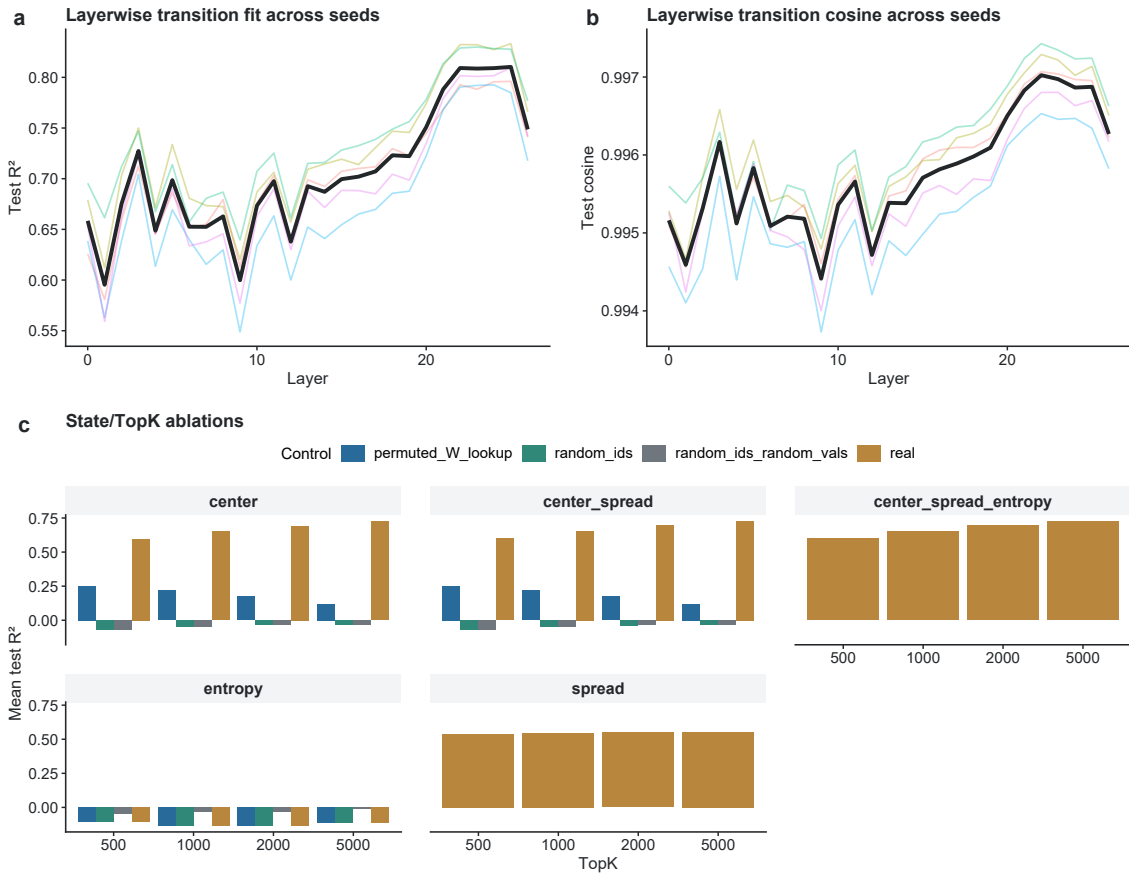
Fig. 2: Robustness of readback-flow organization. **a**, Held-out R^2 distributions across layers and seeds for the fitted transition model and identity-carryover, mean-predictor and shuffled-pair baselines. **b**, Seed-by-layer heat map of held-out transition-model R^2 . **c**, Estimated late macro-transition location τ across TopK values; ribbons show 95% confidence intervals across seeds. **d**, Centre cosine, Jaccard overlap and path agreement under four natural-language perturbation conditions. These statistics have different definitions and should be compared across conditions within a metric, not across metric types. **e**, Mean held-out R^2 in the public-style structured mini-validation; error bars show the reported standard deviation. This panel is a bounded external-style check and not a benchmark-generalization result. See the corresponding Source Data figure and the Source Data manifest.

Supplementary Figure S1 | Prompt inventory and condition ontology
Recorded controlled subsets and checkpoint coverage; exclusions are documented in Methods



Prompt inventory and condition ontology. **a**, Counts in the controlled prompt-condition inventory by recorded condition and mechanism label. **b**, Mechanism composition within each phase label. **c**, Graph-grouped internal-modulation split inventory for the three representative checkpoints. Tokenization and exclusion rules are specified in Supplementary Methods. See Source Data Figs. 9 and 15 and the Source Data manifest.

Supplementary Figure S2 | Full TopK/VIM layer profiles



Full TopK/VIM layer profiles. **a,b**, Layerwise transition-fit R^2 and cosine values across recorded seeds; black curves show seed means. **c**, State/TopK ablations across feature modes and TopK values. Readbacks remain diagnostics rather than causal mechanism variables. See Source Data Figs. 1-3 and the Source Data manifest.

Vocabulary-facing readbacks thus provided a reproducible, layer-resolved observational surface. Static similarity, however, could not distinguish neighbourhoods reached through different transformations. Having the same readout at every layer made those transformations measurable and raised a new question: did successive layers reuse a limited organization? At this point, it was still plausible that the learned output rows exposed structure used during inference.

Did adjacent readback changes reuse a limited repertoire of transformations?

Reproducible readbacks showed which vocabulary entries were visible at each layer. The rearrangement between layers remained unspecified. We represented each readback by its declared centre and spread, then fitted adjacent-layer maps $\hat{F}_l^k : Z_l^k \rightarrow Z_{l+1}^k$. A measurable flow should retain positive held-out fit outside the fitting sample. The maps met this criterion across the stack (Figs. 1b-d and 3b).

The fitted maps suggested a stronger possibility: inference might reuse a small repertoire of layer-to-layer transformations. We clustered their signatures across depth to test this idea. The first path resembled three broad regimes, provisionally labelled input boundary, bulk and output boundary (Fig. 3a). In Qwen, persistent entry into the final regime began at L21-to-L22, pointing to a late macro transition.

Operator clustering revealed a late macro transition, not universal primitives

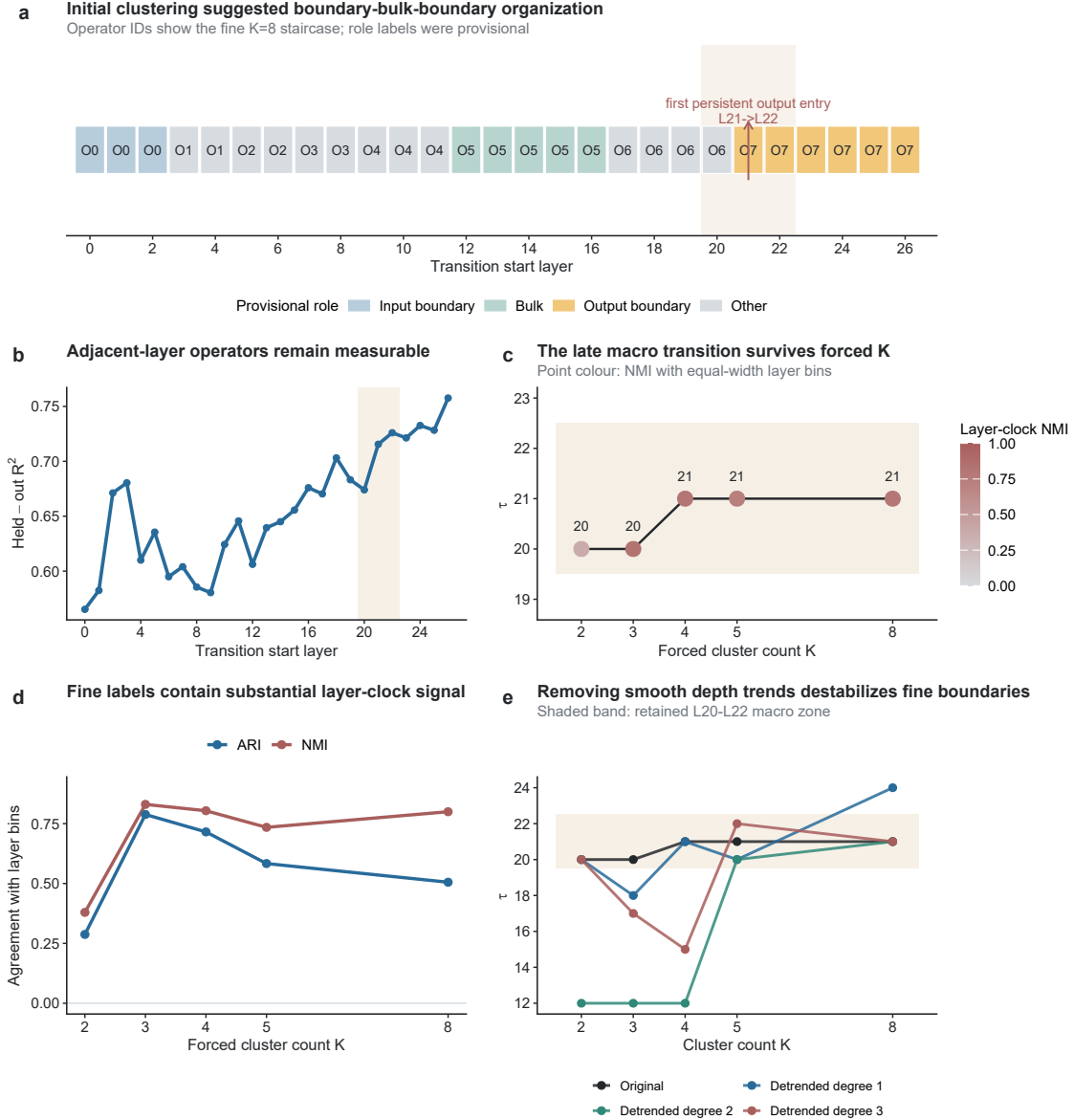
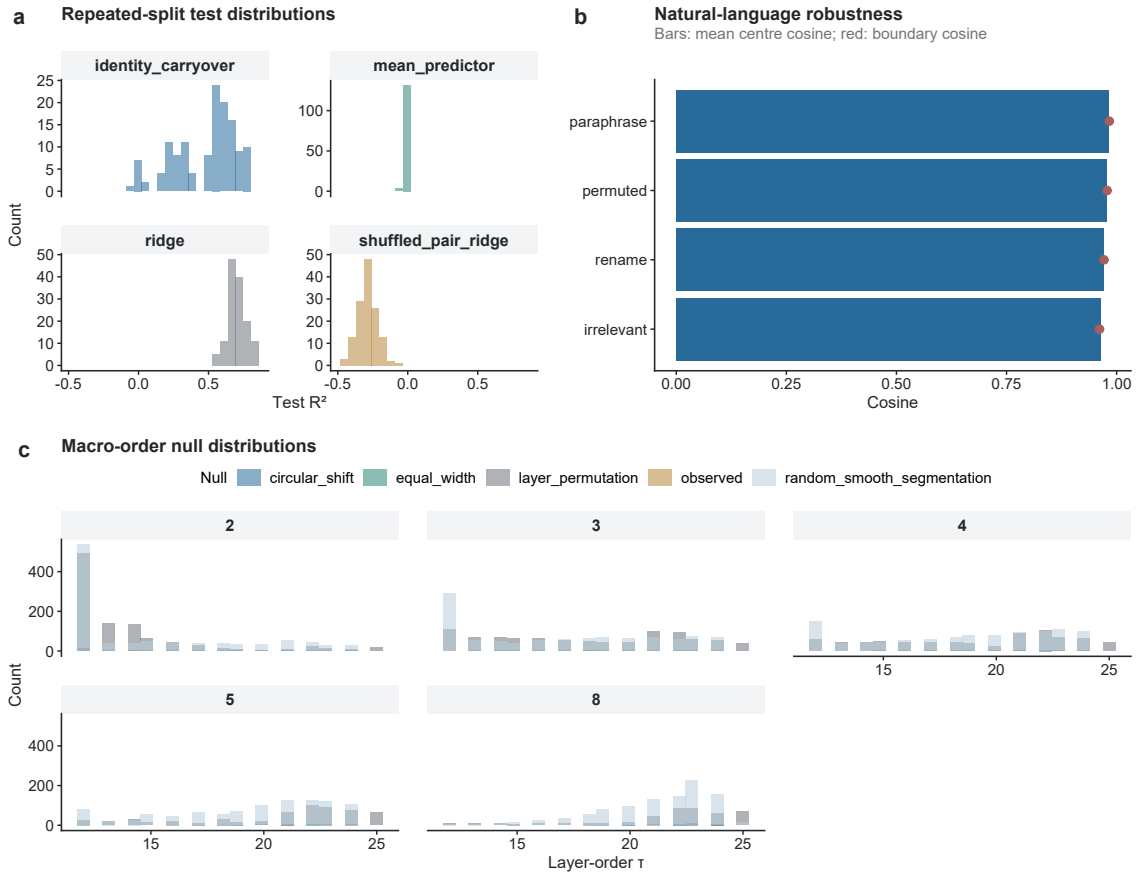


Fig. 3: Operator clustering revealed a late macro transition, not universal primitives. **a**, Initial K=8 clustering path across the 27 adjacent-layer transitions of the Qwen checkpoint. Tiles show operator labels O0-O7; colours show the provisional role assignment used at this discovery stage. The first persistent entry into the output-regime cluster occurred at L21-to-L22. Role assignments were exploratory and are not interpreted as primitive causal operators. **b**, Held-out R^2 of adjacent-layer VIM operator fits across depth. The pale band marks transition start layers 20-22. **c**, Estimated late transition τ under forced cluster counts $K = 2, 3, 4, 5, 8$. The shaded interval marks the retained L20-L22 macro zone; point colour reports normalized mutual information (NMI) between operator labels and equal-width layer bins. **d**, Adjusted Rand index (ARI) and NMI between the original operator labels and equal-width layer bins. High agreement at several cluster resolutions indicates substantial layer-clock structure in the fine labels. **e**, Estimated τ before and after polynomial removal of smooth layer trends. Original estimates remain within L20-L21 across forced K, whereas detrended fine boundaries vary widely. These controls narrow the initial operator-family interpretation to organized readback flow with a robust late macro transition; they do not support a universal catalogue of primitive operators. The analysis used 196 prompt-condition instances, with 147 training and 49 held-out instances for the adjacent-layer fits. Operator clustering used the declared held-out construction. The forced-K values, ARI, NMI and detrended boundary estimates are shown without inferential p-values; they are sensitivity and confound diagnostics. See Source Data Fig. 3 and the Source Data manifest.

Smooth change with depth can also produce discrete-looking clusters. We tested this alternative with forced resolutions $K = 2, 3, 4, 5, 8$, equal-width layer bins, layer-index prediction and depth-detrended reclustering. The fine labels contained substantial layer-clock information (Fig. 3c-e), restricting the three-regime interpretation to an exploratory stage model. A coarser feature remained: the transition estimates were L20, L20, L21, L21 and L21 across the five resolutions (Fig. 3c).

The transition operators remained measurable throughout depth, with held-out R^2 varying systematically and rising in later layers (Fig. 3b). Additional null-segmentation, state/TopK-ablation and relation-preserving analyses showed that the retained macro organization was not reproduced by the tested segmentations and depended on both selected rows and readback features (Supplementary Fig. S3). These checks further separated near-invariant centre geometry from more variable neighbourhood overlap.

Supplementary Figure S3 | Readback robustness and negative examples



Readback robustness and negative examples. **a**, Repeated-split test distributions by baseline. **b**, Natural-language perturbation results. **c**, Macro-order null distributions across K . Weak and negative controls are retained rather than removed. See Source Data Figs. 2-3 and the Source Data manifest.

The layer-clock result changed our interpretation while leaving the observation intact. Readbacks participated in a measurable layerwise flow, and a late macro transition survived. Evidence for a small catalogue of primitive operators was absent. The strongest transitions also occurred at different nominal depths in different checkpoints. Recurrence could no longer be defined by a detailed cluster label or a common layer number, which shifted the comparison to functional windows.

How can the same functional question be compared across checkpoints of different depth?

Cross-checkpoint comparison created a coordinate problem. Nominal layer matching would assume the correspondence under test. Normalized depth would instead assume uniform scaling of function with model depth. We therefore scanned all eligible intervals within each checkpoint and selected windows by the same declared functional objective, without requiring common layer indices or widths. Complete scans showed that each checkpoint and objective selected a different interval (Fig. 4a-c and Supplementary Fig. S4). We call these checkpoint-specific intervals functional windows. Subsequent comparisons used these windows. Selection and evaluation used the same controlled dataset, so best-window values remain descriptive summaries rather than independently confirmed optima. This is functional-role alignment, not evidence for representational homology or a conserved invariant.

Functional roles recur without strict layer or direction identity

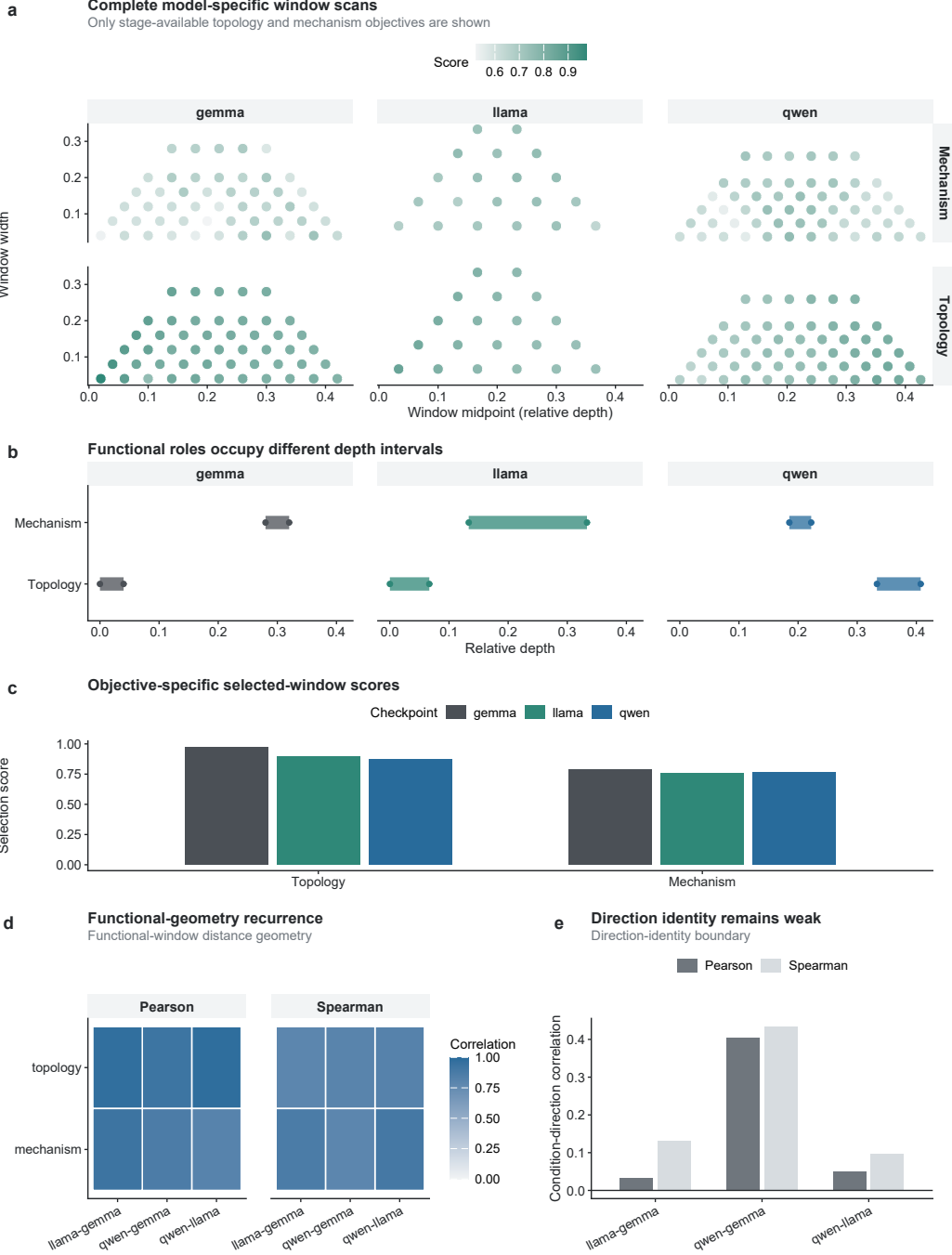
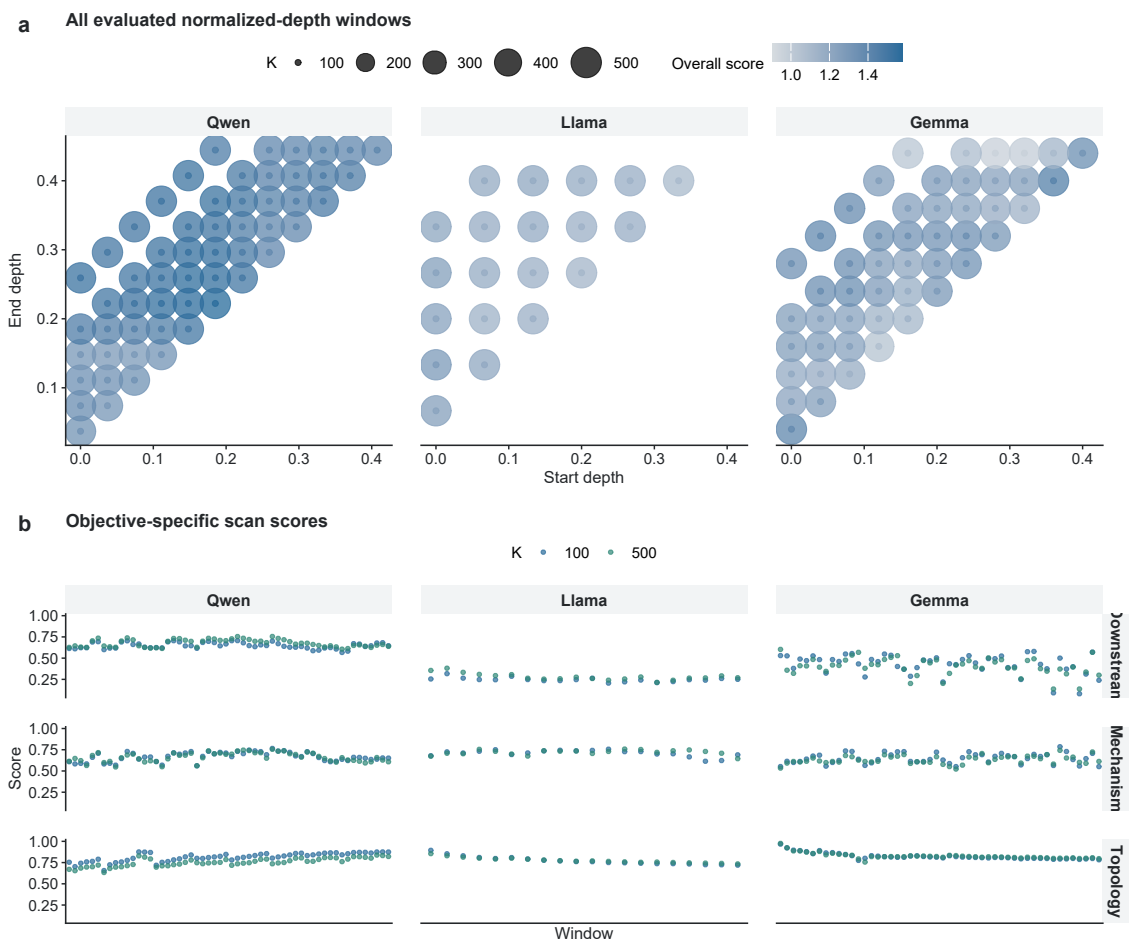


Fig. 4: Functional recurrence without strict layer or direction identity. **a**, Complete model-specific window scans for topology and mechanism objectives in representative Gemma, Llama and Qwen checkpoints. Colour denotes the corresponding selection score. Downstream dominance fields are not used because they become available only after ΔU is introduced later in the discovery chain. **b**, Selected topology and mechanism intervals in relative layer depth. **c**, Objective-specific scores of the selected intervals. **d**, Pairwise recurrence of functional-window distance geometry measured by Pearson and Spearman correlations. **e**, Condition-direction correlations from the separate direction-identity audit. Functional geometry recurred strongly, whereas strict direction identity was weak and heterogeneous. See the corresponding Source Data figure and the Source Data manifest.

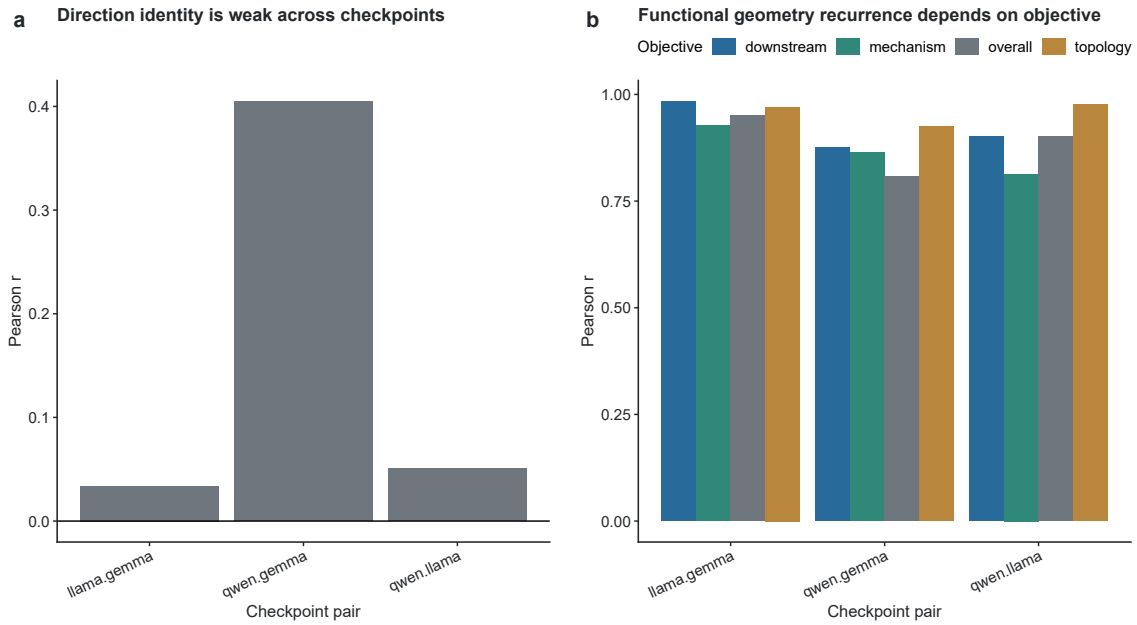
Supplementary Figure S4 | Complete functional-window scans



Complete functional-window scans. **a**, All evaluated windows in normalized checkpoint depth, with colour denoting overall score and point size denoting K. **b**, Objective-specific scores for topology, mechanism-label separation and downstream association. Window recurrence does not imply layer identity. See Source Data Fig. 4 and the Source Data manifest.

Objective-specific window geometry recurred despite this heterogeneity (Fig. 4d). Direction identity was weak. Pairwise correlations of condition-direction distances were weak or checkpoint dependent, whereas functional-geometry correlations were stronger (Fig. 4e and Supplementary Fig. S5). We therefore compared functional roles without assuming common coordinates. The selected windows contained many correlated variables, and their relation to the earlier operator reuse suggested that a continuous accumulation process might admit a compact candidate-relative coordinate.

Supplementary Figure S5 | Direction-isomorphism and non-homology controls
Functional recurrence does not imply coordinate or direction identity



Direction-isomorphism and non-homology controls. **a**, Cross-checkpoint correlations of condition-direction distances. **b**, Recurrence of functional geometry by selection objective. These controls support functional recurrence but reject coordinate or direction identity. See Source Data Fig. 4 and the Source Data manifest.

2 Challenging the readback account

Can many local flow measurements be reduced to one bounded dominance question?

Functional alignment left many local flow measurements to compare. If they reflected a shared candidate-competition process, their clean-relative decision matrix should have a dominant low-rank direction. The first component accounted for 92% of variance in Qwen, 83% in Llama and 97% in Gemma (Fig. 5a,d). Its loadings extended across a continuous late window (Fig. 5b). The oriented scores separated the negative, critical and positive response regimes (Fig. 5c).

Layerwise change concentrates onto a low-rank competition coordinate

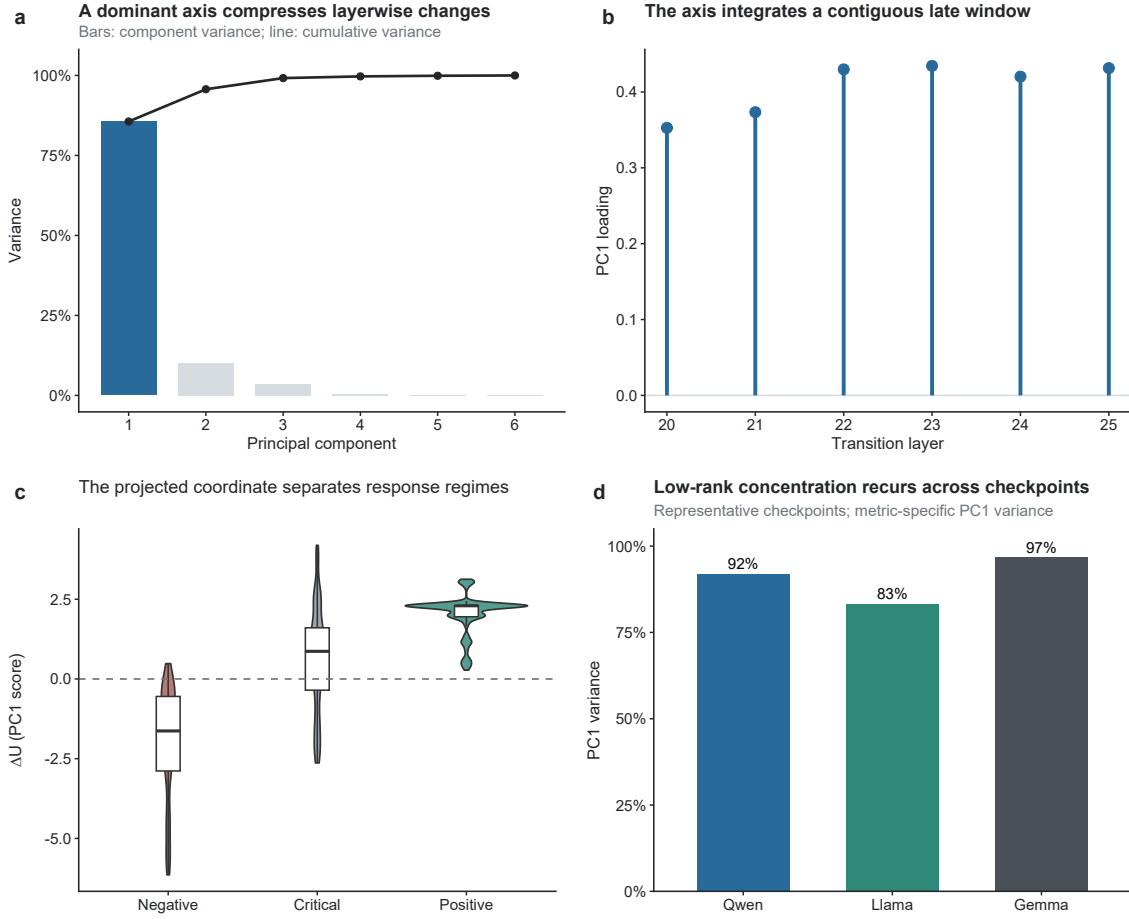
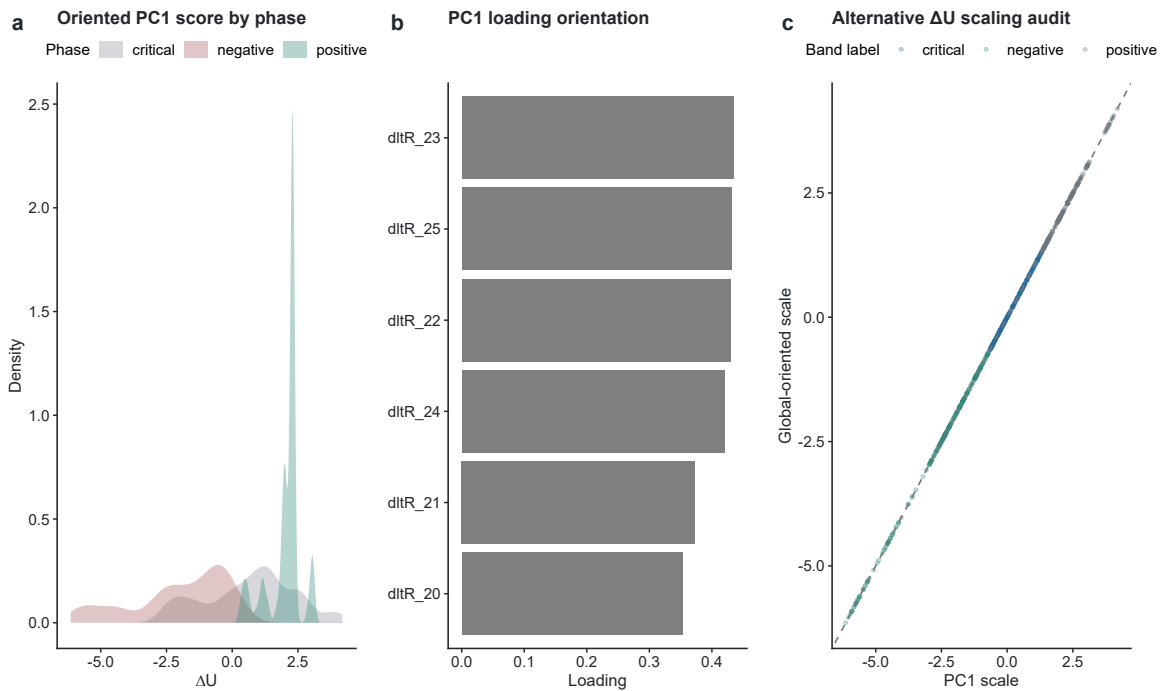


Fig. 5: Layerwise change concentrates onto a low-rank competition coordinate. **a**, PCA spectrum of six late-window layer-to-layer readback changes in the original low-rank dominance analysis. Bars show variance explained by each component; the line shows cumulative variance. **b**, Loadings of transitions L20-L25 on PC1. **c**, Distribution of the oriented PC1 coordinate, subsequently denoted ΔU , across negative, critical and positive response regimes ($n = 768$ prompt-condition observations). Violin shapes show empirical density and boxes show medians and interquartile ranges. **d**, Fraction of variance concentrated on PC1 in representative Qwen, Llama and Gemma checkpoints. Cross-checkpoint values are metric-specific and do not imply a shared physical axis. See Source Data Fig. 5 and Source Data manifest.

We defined the signed candidate-path separation as ΔU , a low-rank coordinate that summarizes relative support for the leading and competing candidate trajectories within a declared window. It is a bounded dominance observable. Let D denote the clean-relative decision matrix over the permitted decision-window features. We oriented its leading principal direction towards greater support for the leading trajectory. The competing trajectory supplied the reference. Final answer labels were excluded as an independent validation target. Supplementary Fig. S6 reports the loadings, sign audit and alternative scaling.

Supplementary Figure S6 | ΔU construction and orientation audit

Sign is oriented by the controlled leading-versus-competing contrast; it is not an independent target-free label



ΔU construction and orientation audit. **a**, Oriented PC1 score distributions by controlled phase. **b**, PC1 loadings on the decision-window contrast features. **c**, PC1 and global-oriented scalings. The sign of ΔU is oriented by the controlled leading-versus-competing contrast; ΔU remains a bounded observable and is not an independently target-free label. See Source Data Figs. 5-6 and the Source Data manifest.

The ΔU distributions formed checkpoint-specific competition bands. Negative, critical and positive conditions overlapped in the full feature space but were ordered along the oriented coordinate (Fig. 6a,b). Band occupancy varied by condition, and a band-only classifier reached macro-F1 0.61 (Fig. 6c,d). Because the coordinate and condition contrast shared the controlled candidate-pair design, this agreement was partly expected. We treated it as a construction check, not independent evidence for a mechanism, correctness variable or universal order parameter. The more demanding test was whether related structure appeared before the decision window used to define ΔU .

ΔU exposes a bounded competition band

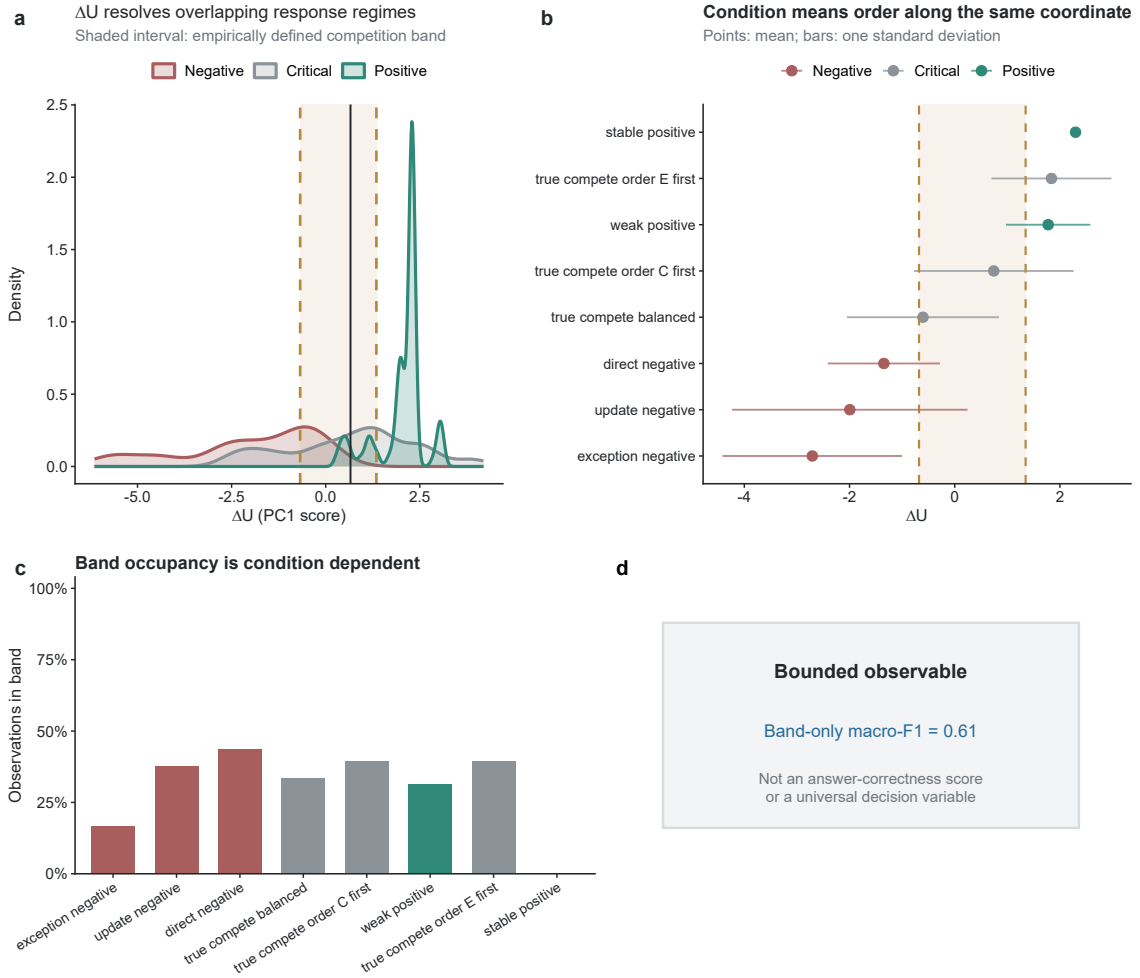


Fig. 6: ΔU exposes a bounded competition band. **a**, Empirical ΔU distributions for negative, critical and positive response regimes. The shaded interval is the empirically defined competition band $[U_-, U_+]$; the solid line marks its critical centre. **b**, Condition-level means and one standard deviation, ordered along ΔU . **c**, Fraction of observations from each condition falling inside the band. **d**, Boundary summary. A classifier using the band midpoint alone achieved macro-F1 = 0.61 in this controlled dataset. ΔU is a bounded observable, not an answer-correctness score or a universal decision variable. See Source Data Fig. 6 and Source Data manifest.

Does dominance-relevant structure exist before the decision window?

The compactness of ΔU created an immediate circularity risk. A predictor using the decision window could appear informative simply because it reread the features used to construct the target. We therefore set a threat-driven criterion: a precursor interpretation required some association to survive both temporal exclusion of the decision window and removal of R-derived candidate-pair features. We first retained only predecision transitions, then restricted the predictors to non-R vocabulary-neighbourhood features. All rows sharing a relation graph were kept in the same split, and uncertainty was estimated by resampling relation-graph groups with model refitting.

Association with later ΔU remained after the predecision restriction in all three checkpoints, although its magnitude differed (Fig. 7a). Non-R features also retained positive association and mechanism-label separation (Fig. 7b). Grouped bootstraps gave strong Qwen and Gemma correlations and a weaker, less certain Llama result (Fig. 7c,d and Supplementary Fig. S7). These controls reduced direct leakage; causal status for the vocabulary readback remained unsupported. They left a more basic question unresolved: did the preserved geometry depend on learned vocabulary-row identity?

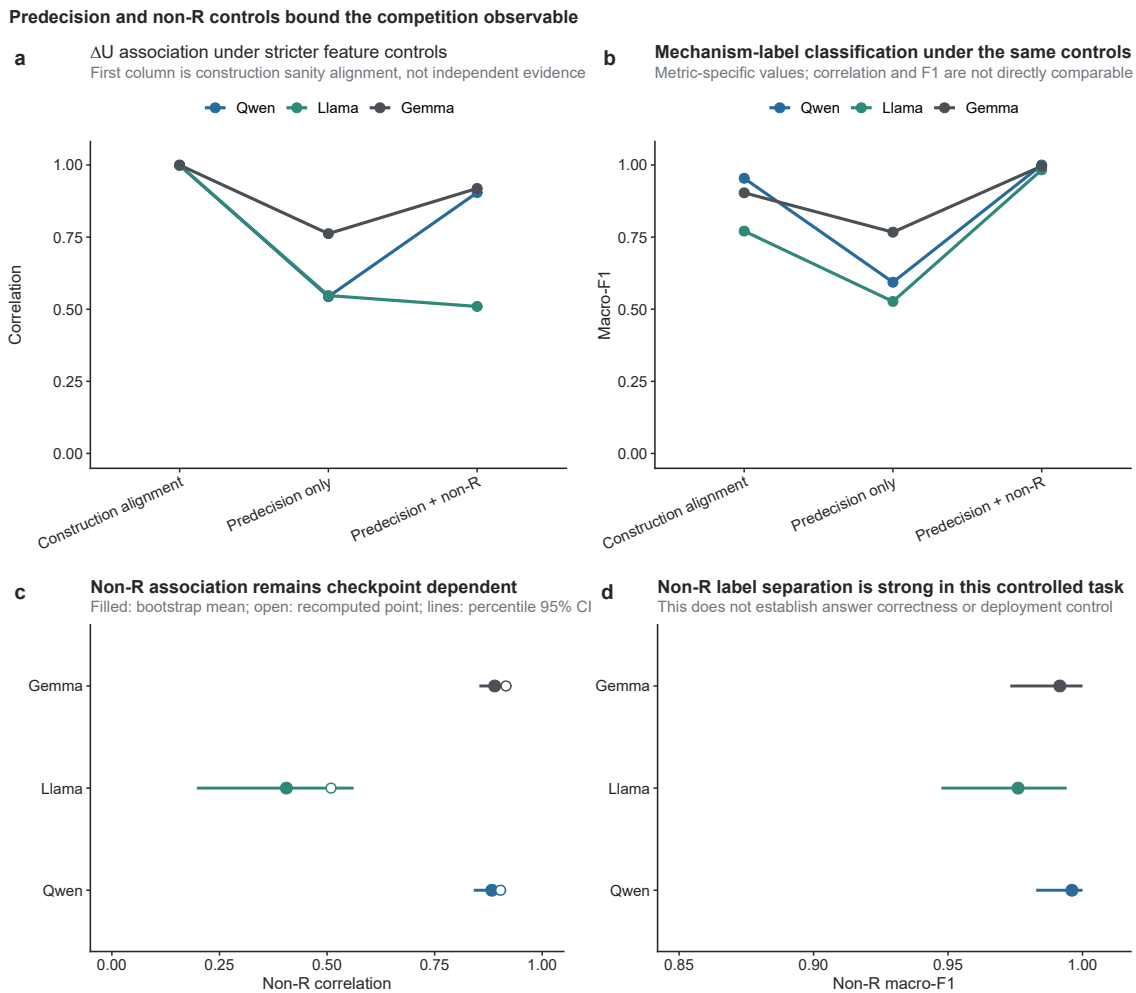
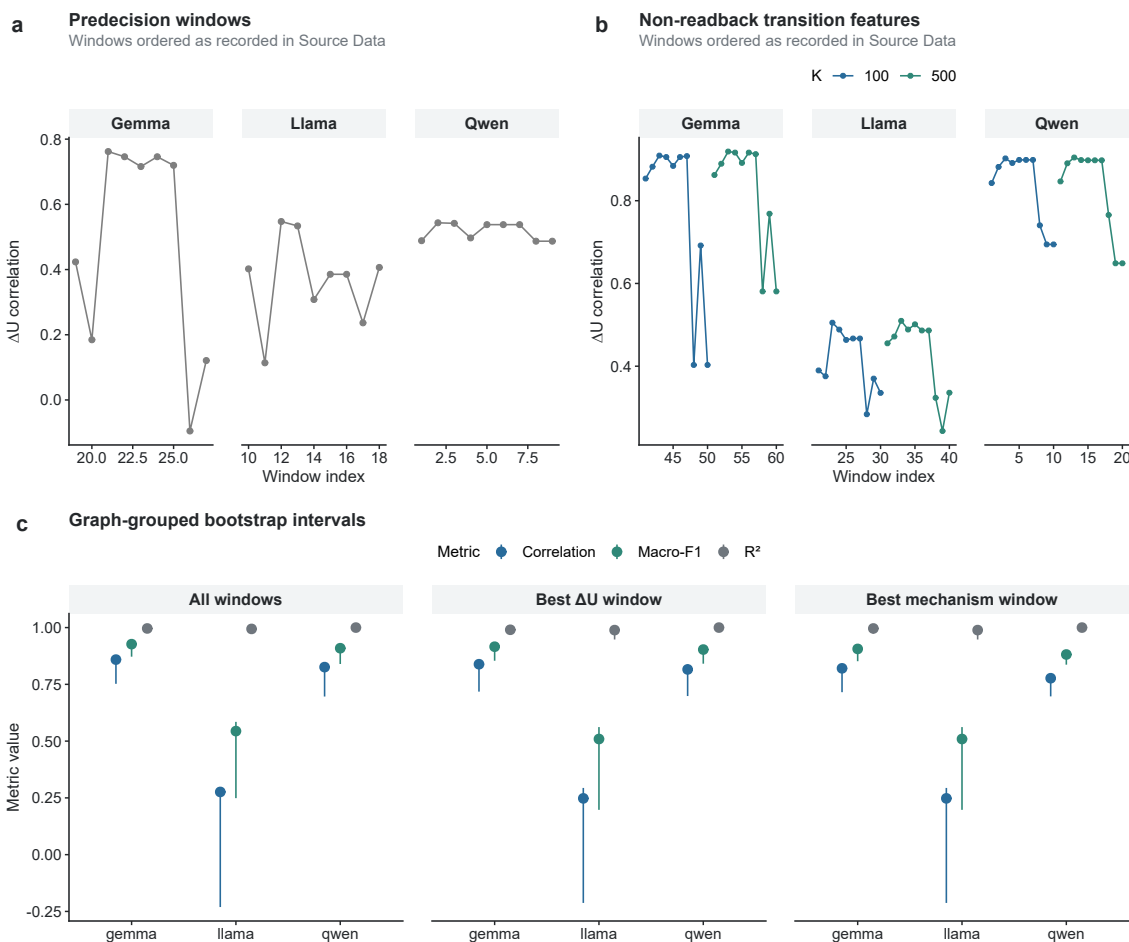


Fig. 7: Predecision and non-R controls bound the competition observable. **a**, Association with ΔU across three feature constructions: unrestricted construction alignment, features restricted to transitions before the decision window, and predecision vocabulary-neighbourhood-flow features excluding R. The unrestricted value is a construction sanity alignment and is not independent mechanistic evidence. **b**, Mechanism-label macro-F1 under the same control ladder. Correlation and macro-F1 are metric-specific and are not directly comparable. **c**, Non-R correlation for the best ΔU window. Filled points are grouped-bootstrap means, open points are recomputed full-sample estimates, and lines are percentile 95% confidence intervals (200 successful resamples per checkpoint). **d**, Non-R mechanism-label macro-F1 for the best mechanism window, shown as bootstrap means and percentile 95% intervals. Strong controlled-task label separation does not establish answer correctness or deployment control. See Source Data Fig. 7 and Source Data manifest.

Supplementary Figure S7 | Predecision leakage and bootstrap audit



Predecision leakage and bootstrap audit. **a**, Predecision hidden-flow windows, all excluding the decision window. **b**, Non-readback neighbourhood-transition windows. Window indices preserve Source Data order. **c**, Graph-grouped bootstrap intervals for correlation, mechanism macro-F1 and R2. These controls reduce but do not erase dependence on the controlled answer-pair construction. See Source Data Fig. 7 and the Source Data manifest.

Is the observed geometry carried by learned vocabulary identity?

The early readback results were compatible with a strong explanation: learned vocabulary rows might carry the geometry that made the layerwise patterns informative. This interpretation supplied a discriminating prediction. If learned row identity was essential, row permutation or replacement by matched random projections should sharply reduce preservation. We tested the learned output projection against three controls. Row permutation preserved the row set while removing identity. The Gaussian control replaced learned rows within a matched projection-selection pipeline. Random-K disrupted conditional selection.

Learned-W and row-permuted-W readbacks preserved hidden geometry almost identically (Fig. 8b,d). Gaussian projections often preserved more, and the pattern persisted across sampled layers (Fig. 8b,c). Random-K selection sharply reduced preservation (Fig. 8e). These controls reject learned output-row identity as the source of the preserved geometry. They instead identify projection and conditional selection as the operations that retain it. TopK/VIM therefore remains a useful diagnostic readback, but not a privileged mechanism coordinate.

Projection-selection controls falsify the learned-row-identity explanation

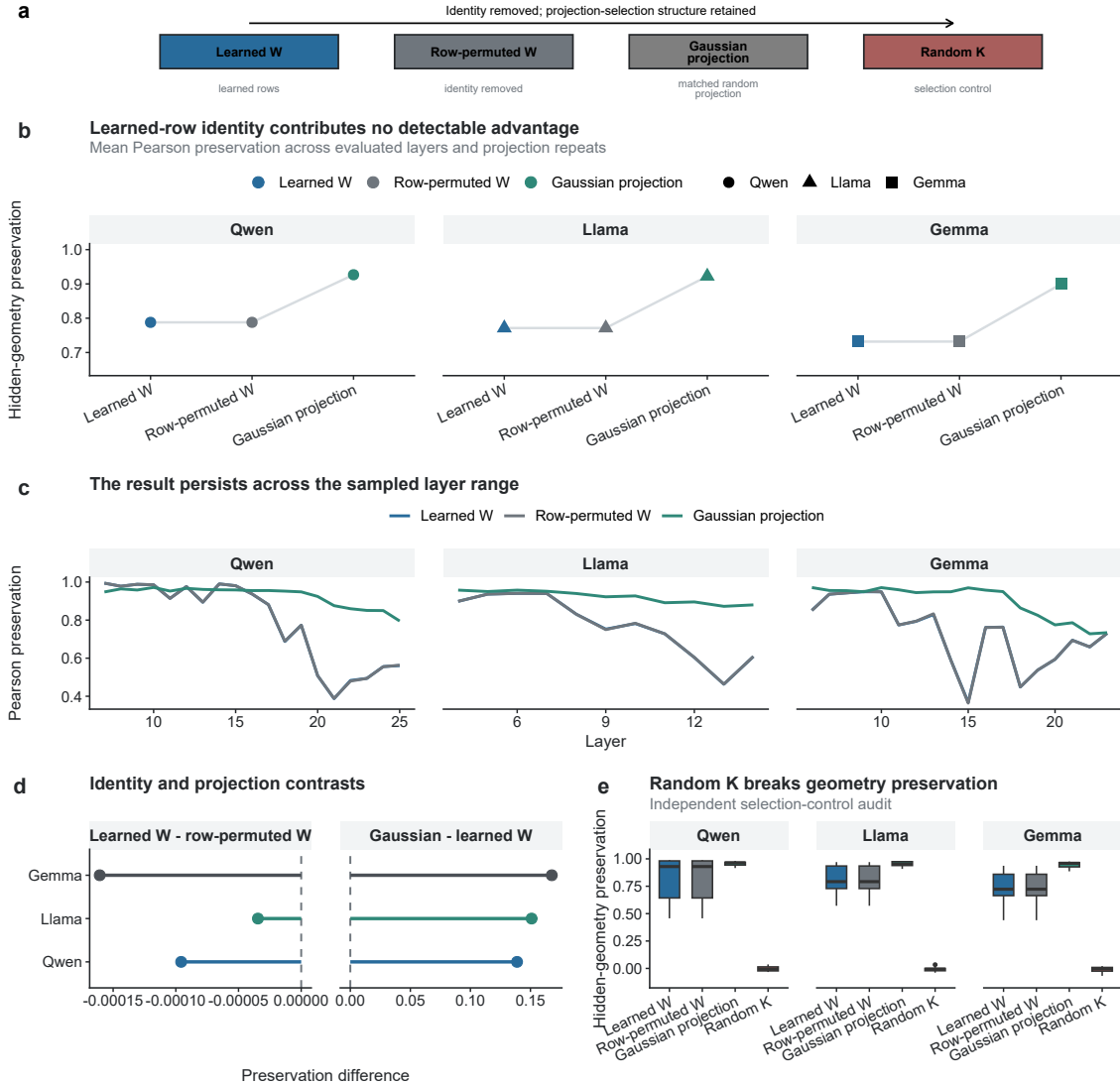


Fig. 8: Projection-selection controls falsify the learned-row-identity explanation. **a**, Control logic. The learned output projection was compared with a row-permuted version that removes vocabulary-row identity while preserving the row set, and with a matched Gaussian projection. Random-K selection was evaluated in a separate cross-checkpoint selection-control audit. **b**, Mean Pearson preservation of pairwise hidden-state geometry across evaluated layers for representative Qwen, Llama and Gemma checkpoints. Learned and row-permuted output projections are indistinguishable at the displayed scale, whereas the Gaussian projection is stronger. **c**, Layer-resolved preservation averaged over projection repeats. The learned-W and row-permuted-W curves overlap almost exactly; apparent disappearance of the learned-W curve reflects this overlap rather than missing data. **d**, Model-level contrasts. Learned-W minus row-permuted-W preservation is approximately zero, while Gaussian-minus-learned-W preservation is positive in all three checkpoints. **e**, Independent controlled proxy-audit dataset selection control. Learned-W, row-permuted-W and Gaussian TopK constructions preserve hidden geometry substantially better than random-K selection. Panel e is not pooled with the projection-selection geometry control projection repeats in panels b-d. These controls reject learned output-row identity as the explanation for readback geometry while retaining vocabulary readbacks as non-random diagnostic observables. Gaussian projection performance does not establish a superior mechanism coordinate. See Source Data Fig. 8 and Source Data manifest.

The control removed the basis for privileging vocabulary-facing coordinates. We next compared them directly with variables derived from hidden states.

Which object is closer to the process than the vocabulary readback?

Once learned row identity failed, vocabulary-facing variables could no longer be treated as privileged mechanism coordinates. The failure forced a direct comparative question: did variables derived from hidden states and their ordered changes provide stronger coordinates for later dominance, phase and mechanism labels than the projection-selection readback? The proxy-ranking audit tested this comparison using 960 prompt-condition instances from 80 relation-graph groups and 12 conditions, evaluated by five-fold grouped cross-validation (Fig. 9a). This dataset is distinct from the 96-prompt controlled subset used in the projection-selection audit.

Hidden-state proxies outrank vocabulary readbacks in the controlled proxy audit

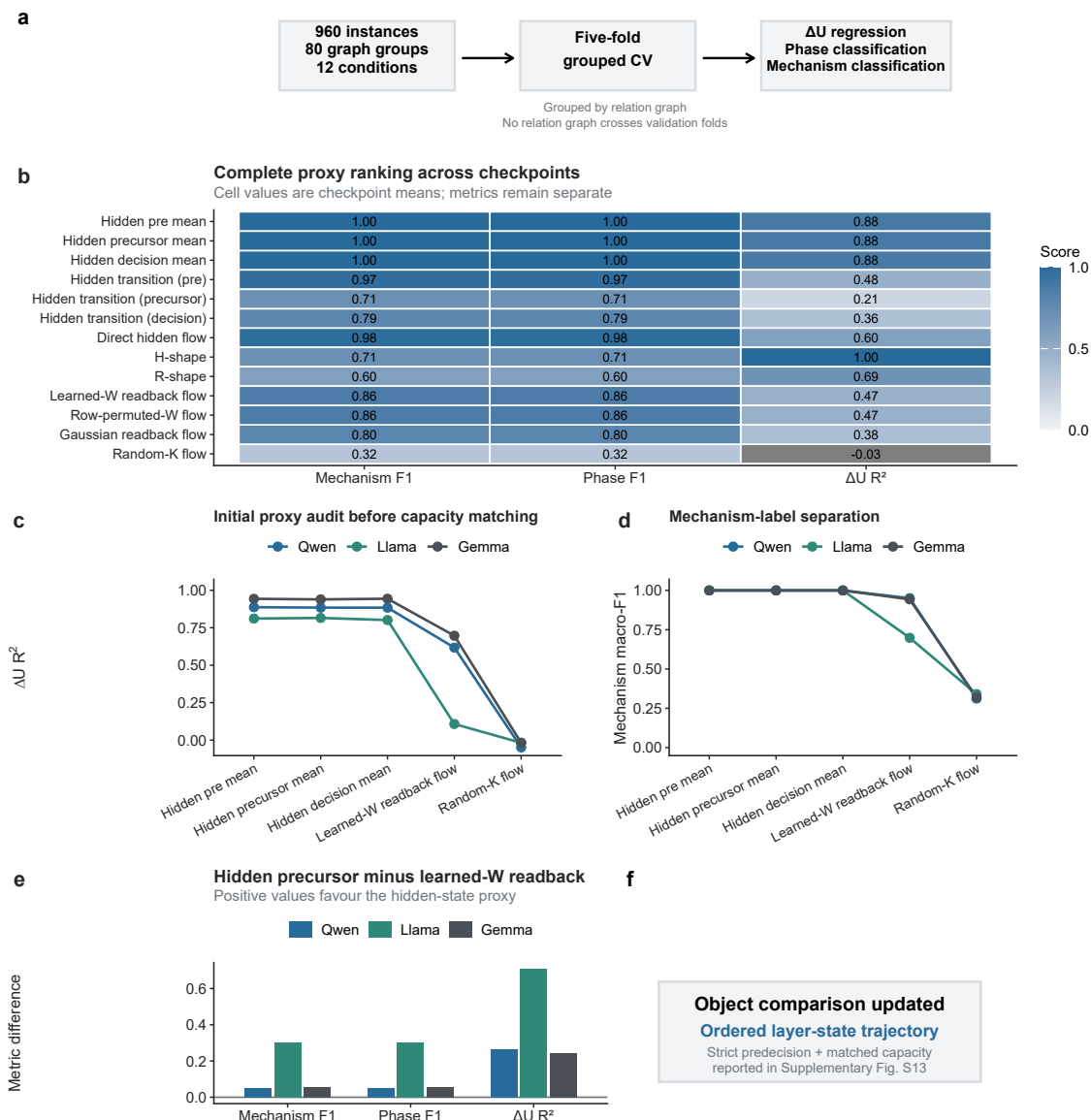


Fig. 9: Hidden-state proxies outrank vocabulary readbacks in the controlled proxy audit. **a**, Proxy-audit design for 960 prompt-condition instances from 80 relation-graph groups and 12 conditions, evaluated by five-fold grouped cross-validation. **b**, Initial proxy ranking across representative Qwen, Llama and Gemma checkpoints; cells report $\Delta U R^2$, phase macro-F1 and mechanism macro-F1 separately. **c**, Checkpoint-specific ΔU regression for selected hidden-state, learned-W readback and random-K proxies. **d**, Mechanism-label macro-F1 for the same proxies. **e**, Hidden-precursor minus learned-W performance for each metric. **f**, Object-update decision. Hidden-state variables lead this initial audit; the strict-predecision matched-capacity test in Supplementary Fig. S13 retains their mechanism-label advantage while showing checkpoint- and dimension-dependent ΔU results. H-shape is a compressed descriptor, not complete trajectory recovery. See Source Data Fig. 9 and Source Data manifest.

Hidden predecision, precursor and decision summaries ranked above learned-W, row-permuted-W and Gaussian readback flows for later ΔU , phase and mechanism labels in the tested setting (Fig. 9b-d). The hidden precursor advantage over learned-W readback was positive across the three evaluated metrics and all representative checkpoints, with the largest ΔU advantage in Llama (Fig. 9e). A second audit asked whether each proxy improved closure of a future-state model. Fold-local dimensionality reduction was fitted inside each grouped training fold. Hidden and trajectory-shape variables produced larger future-state and mechanism-label gains than vocabulary readbacks, whereas random-K was near zero or negative (Fig. 10b-d). A composite gain summary gave the same ordering but was retained only as a secondary view because its component metrics are not directly comparable (Fig. 10e).

Fold-local proxy reduction preserves the hidden-state closure advantage

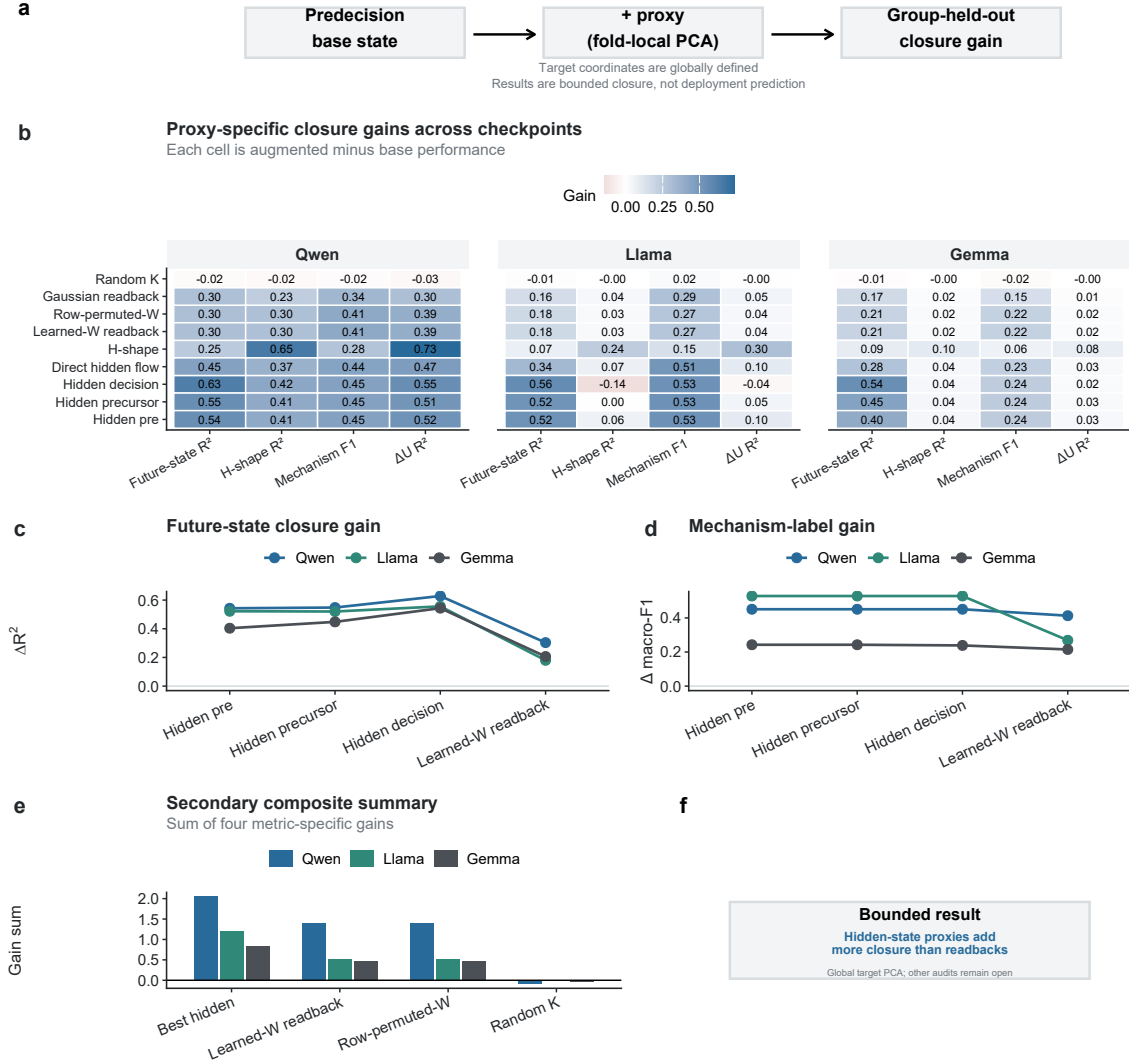
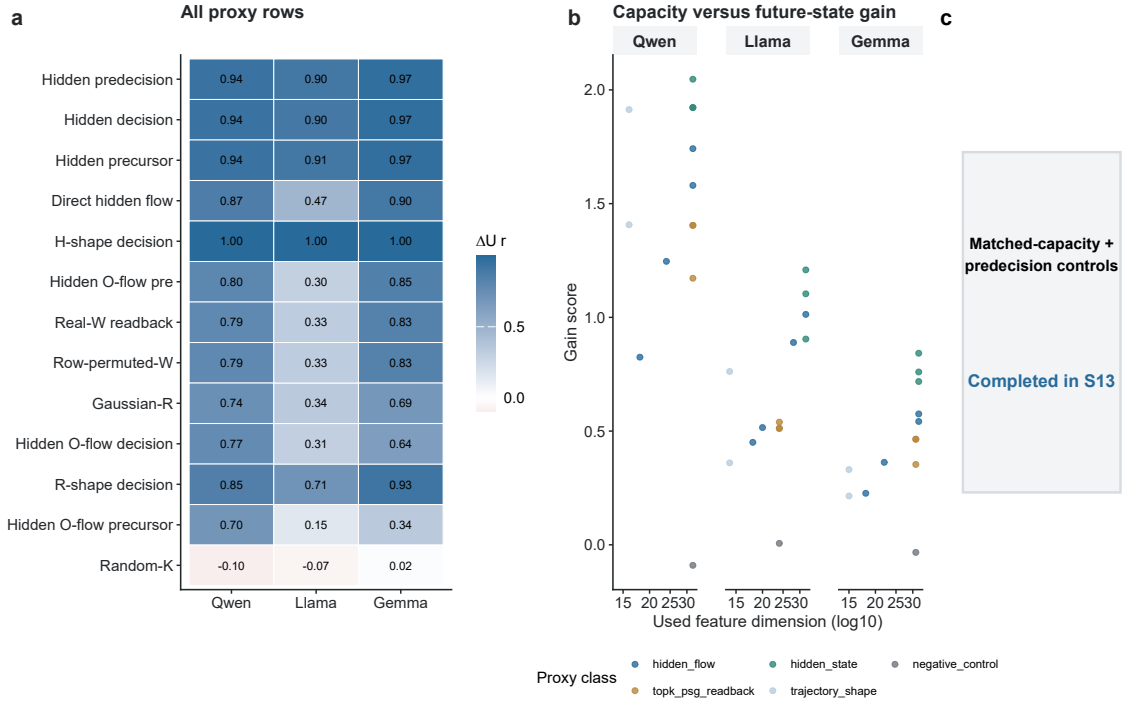


Fig. 10: Fold-local proxy reduction preserves the hidden-state closure advantage. **a**, Corrected closure-gain design. Each proxy augments a low-dimensional predecision base state. For proxies exceeding the dimensional cap, PCA is fitted only on the training groups of each five-fold relation-graph-grouped cross-validation split and then applied to held-out groups. **b**, Augmented-minus-base gains for future-state R^2 , H-shape R^2 , mechanism macro-F1 and $\Delta U R^2$ across representative checkpoints. **c,d**, Checkpoint-specific gains for selected hidden-state summaries and learned-W readback flow. **e**, Secondary composite view, defined as the sum of the four displayed metric-specific gains. The composite is not treated as a single physical quantity. Learned-W and row-permuted-W readback gains remain nearly equal (maximum score difference 0.001078), while random-K gains are near zero or negative. **f**, Boundary. Hidden/state proxies add more group-held-out closure than vocabulary readbacks in this corrected analysis, but the future hidden-state target PCA and ΔU orientation are globally defined descriptive coordinates. The result therefore does not establish deployment-style future prediction. Supplementary Fig. S13 separately reports the completed strict-predecision matched-capacity audit. See Source Data Fig. 10 and Source Data manifest.

Supplementary Figure S8 | Proxy-rank and capacity audit



proxy-coordinate proxy-rank and capacity audit. **a**, Full proxy-by-checkpoint ΔU correlation matrix. **b**, Future-state gain score against retained feature dimension. **c**, Design of the strict-predecision matched-capacity audit completed in Supplementary Fig. S13. The initial ranking is not interpreted as capacity matched. See Source Data Figs. 9-10 and the Source Data manifest.

These grouped comparisons establish the ordered layer-state trajectory as the primary empirical object for the remaining mechanism analyses. We define $T_{m,p} = (h_{m,p,1}, \dots, h_{m,p,L_m})$ as the ordered hidden states for checkpoint m and prompt p (Fig. 11a). From this point onward, mechanism claims refer to this ordered hidden-state trajectory; vocabulary readbacks are retained only as diagnostic projections or comparison coordinates. Checkpoint-specific projections showed organized centroid paths and depth-dependent separation, with much of the variation concentrated in high-variance dimensions (Fig. 11b-e). These projections orient the reader to the trajectory; the quantitative tests use the declared high-dimensional coordinates.

High-dimensional layer-state trajectories organize the retained observables

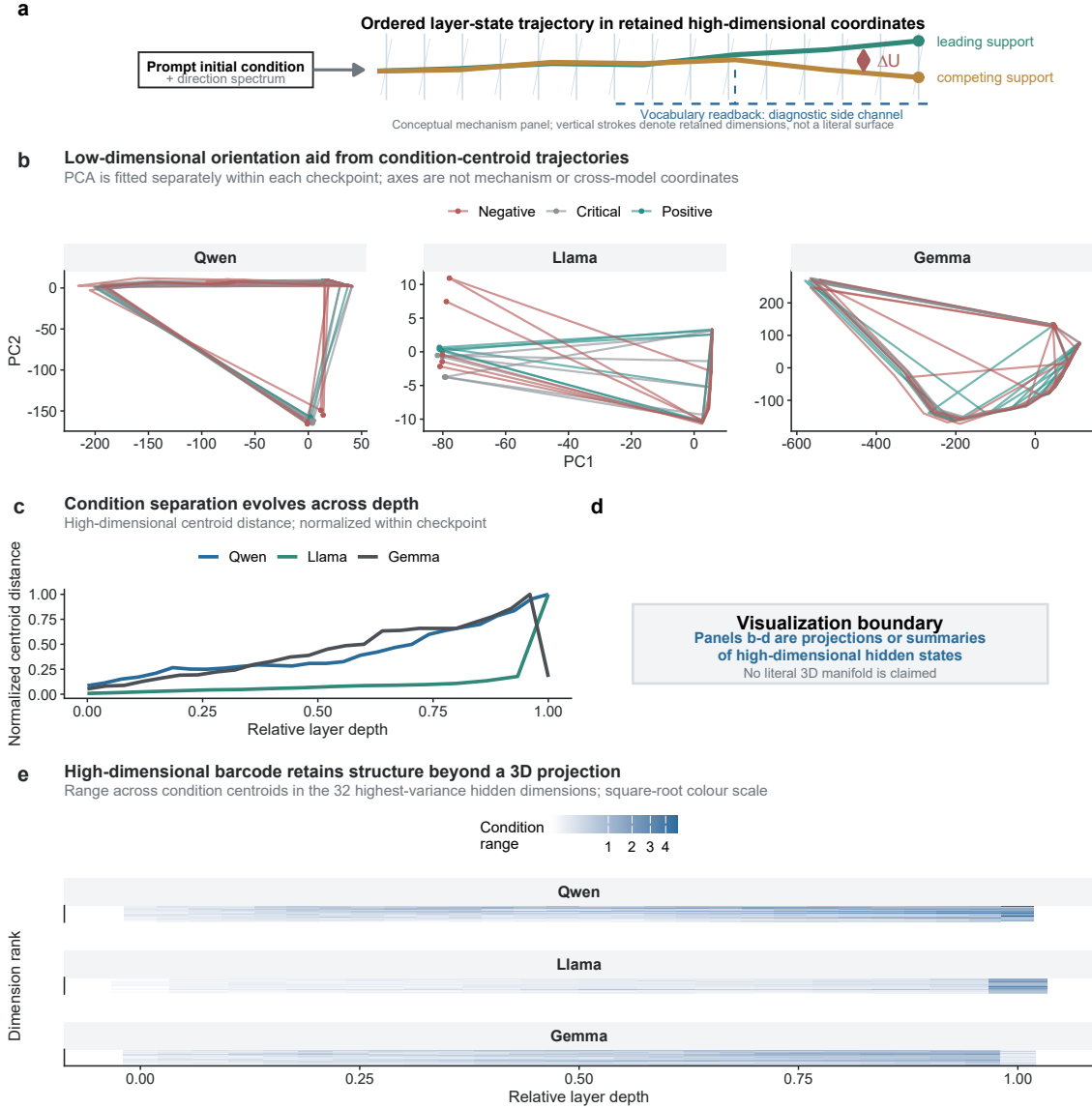
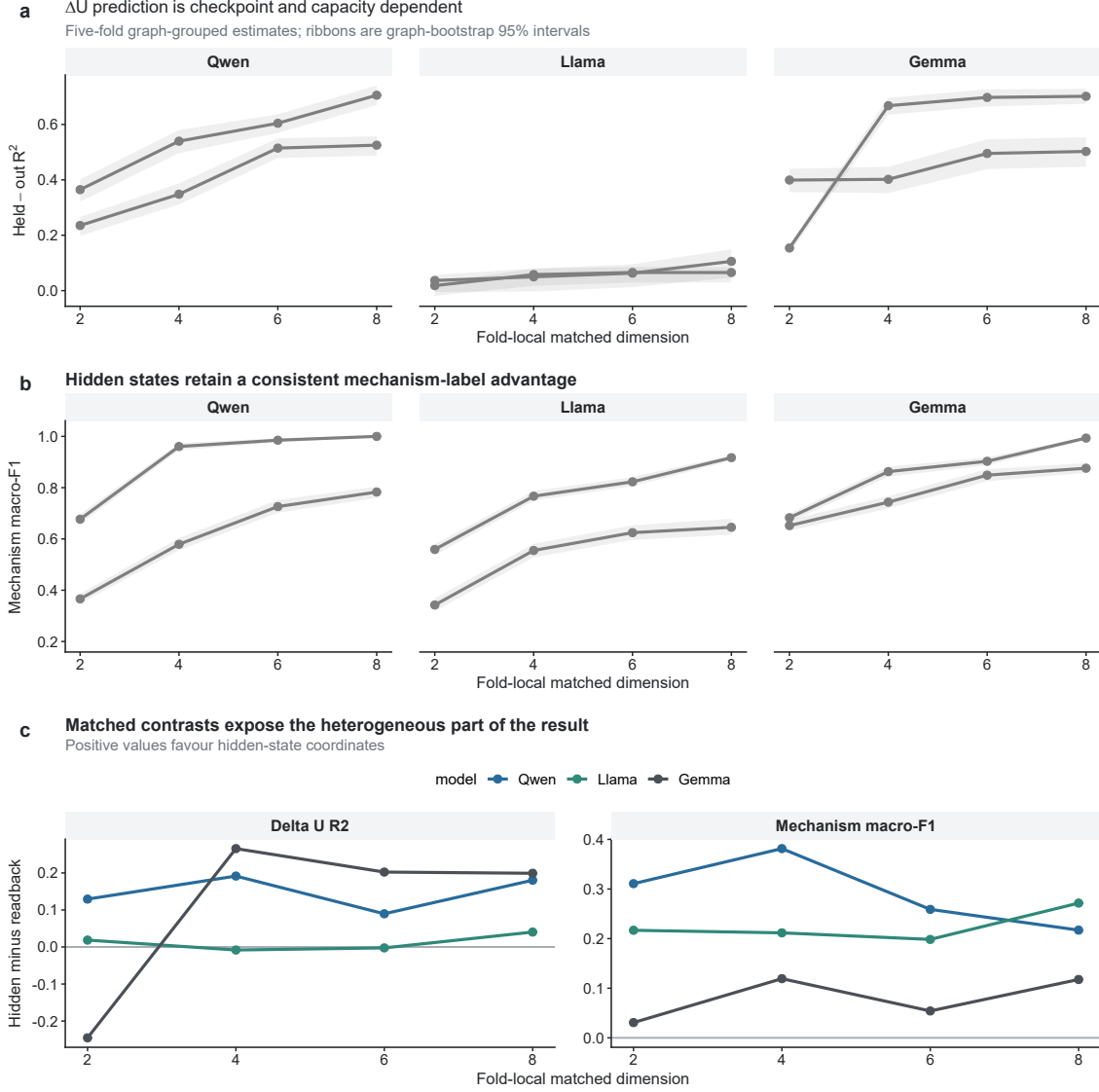


Fig. 11: High-dimensional layer-state trajectories organize the retained observables. **a**, Conceptual high-dimensional mechanism view. A prompt supplies the initial condition and direction spectrum; the ordered layer-state sequence develops in retained high-dimensional coordinates, while leading and competing support separate along the bounded observable ΔU . Vertical strokes denote retained dimensions rather than a literal surface. Vocabulary-facing readback is shown as a diagnostic side channel. The ordered hidden-state sequence remains the primary empirical object; transition coordinates, ΔU and structured residuals are derived diagnostics. **b**, Low-dimensional orientation aid obtained from the full hidden-state arrays for 960 prompt-condition instances. PCA was fitted separately within each checkpoint over all condition-by-layer centroids. Lines connect successive layers and are coloured by controlled phase label. Axes are checkpoint-specific projection coordinates, not mechanism coordinates, and cannot be compared numerically across panels. **c**, Mean pairwise Euclidean distance among high-dimensional condition centroids at each layer, normalized by the within-checkpoint maximum. Late-layer decreases for Qwen and Gemma are present in the extracted data and are not missing segments. **d**, Visualization boundary: panels b-e are projections or summaries of high-dimensional hidden states and do not claim that the underlying object is literally a three-dimensional manifold. **e**, Range across condition centroids in the 32 hidden dimensions with highest variance in each checkpoint. Colour uses a square-root scale to reveal non-negative range structure without altering the underlying values. Dimension ranks and PCA axes are selected separately within each checkpoint. See Source Data Fig. 11 and Source Data manifest.

Dimensional capacity and proximity to the target could still explain that ranking. We therefore repeated the comparison using only predecision layers and reduced every proxy to 2, 4, 6 or 8 dimensions inside each grouped training fold. Hidden coordinates retained higher mechanism-label macro-F1 at every checkpoint and dimension (Supplementary Fig. S13b,c). Their ΔU advantage was not universal: it was positive throughout Qwen, near zero at intermediate dimensions in Llama, and reversed at two dimensions in Gemma before becoming positive at 4-8 dimensions (Supplementary Fig. S13a,c). The matched audit supports hidden trajectories for controlled mechanism-label organization while showing that dominance prediction depends on checkpoint and retained capacity. We next used transition closure to examine how successive states were related.

Strict-predecision matched-capacity audit



Strict-predecision matched-capacity object audit. **a**, Held-out ΔU R^2 for hidden-state and vocabulary-readback coordinates after both were reduced to 2, 4, 6 or 8 dimensions inside each graph-grouped training fold. Only transitions before the declared decision window entered either proxy. Ribbons show graph-bootstrap 95% intervals. The hidden-state advantage is checkpoint and capacity dependent: it is positive throughout Qwen, near zero at intermediate dimensions in Llama, and reverses at two dimensions in Gemma. **b**, Mechanism-label macro-F1 under the same strict-predecision, matched-capacity construction. Hidden-state coordinates are higher at every tested checkpoint and dimension. **c**, Hidden-minus-readback contrasts, shown without aggregation across metric types. Positive values favour hidden-state coordinates. The result supports a stable mechanism-label advantage but not a universal ΔU advantage. Five-fold cross-validation kept relation graphs intact; uncertainty was estimated by 500 graph-level bootstrap resamples of out-of-fold predictions. See Source Data Supplementary Fig. S13 and the Source Data manifest.

3 Following the ordered hidden-state trajectory

Can layer-to-layer evolution be described without claiming prediction?

A trajectory object by itself leaves the relation between successive states unspecified. We therefore asked whether compact coordinates of the observed transition could close more of the adjacent-state relation than state descriptors alone. The crucial distinction was whether such closure remained descriptive or could be recovered from information available before the transition. The six-feature H-shape descriptor, which records local velocity changes, reversals and centre backtracking, modestly increased closure over state alone, whereas continuous latent transition coordinates O_l^{cont} produced a much larger increase. Macro R^2 rose from 0.820 for the state model and 0.824 for state plus H-shape to 0.980 for state plus transition coordinates and 0.993 for a bilinear state-transition model (Fig. 12b). Removing O_l^{cont} caused the largest ablation drop (Fig. 12c).

Continuous transition coordinates provide descriptive closure but weak predictive retention

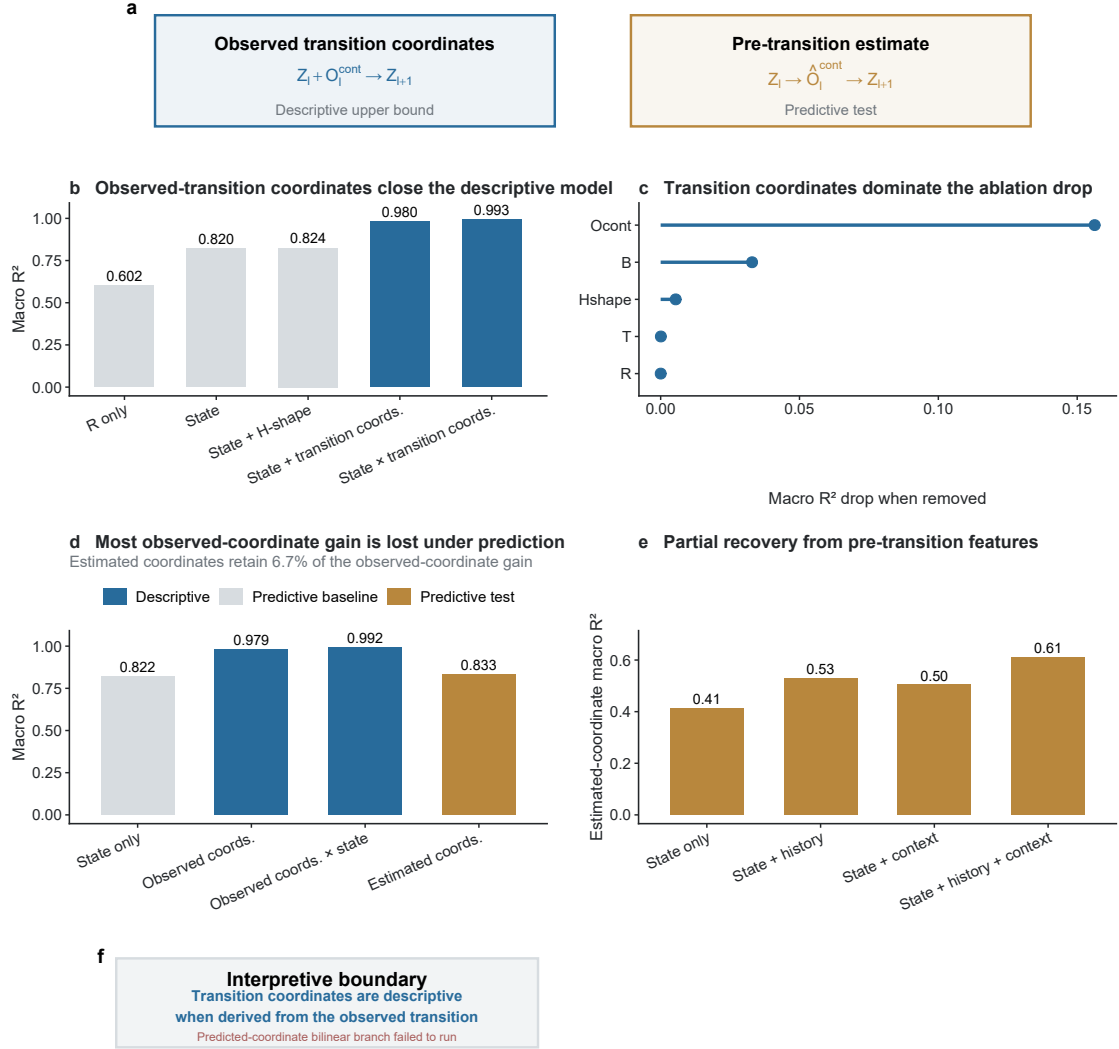
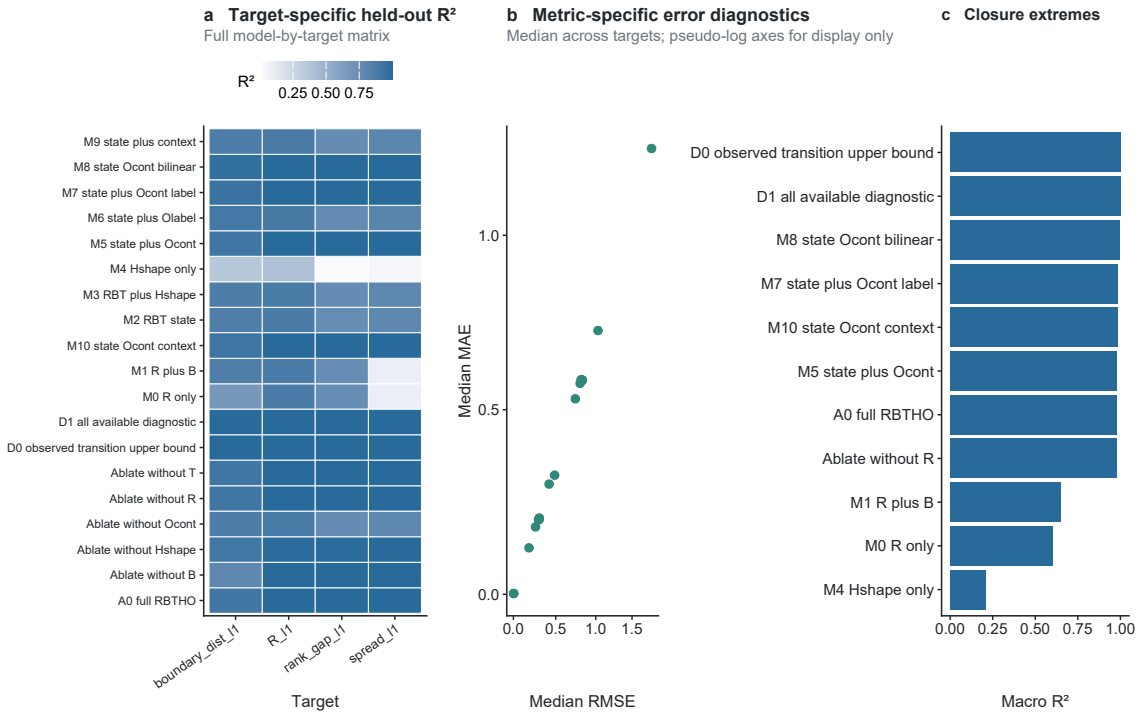


Fig. 12: Continuous latent transition coordinates provide descriptive closure but weak predictive retention. **a**, Information boundary. Observed continuous transition coordinates O_t^{cont} are derived from the realized transition and provide a descriptive upper bound. Estimated coordinates $\hat{O}_t^{cont}$ use pre-transition state/history/context and constitute the predictive test. **b**, one-step transition-closure audit model ladder. Adding observed O_t^{cont} raises grouped macro- R^2 from 0.824 for state plus H-shape to 0.980; state-coordinate interactions reach 0.993. **c**, Leave-component-out analysis. Removing O_t^{cont} produces the largest macro- R^2 drop. **d**, transition-coordinate-7F comparison. State-only prediction reaches 0.822, observed coordinates reach 0.979, and the best model using pre-transition estimated coordinates reaches 0.833. Estimated coordinates therefore retain 6.7% of the observed-coordinate gain under the implemented test. **e**, Predictability of O_t^{cont} from increasingly rich pre-transition feature sets; the best macro- R^2 is 0.61. **f**, Interpretation boundary. Adjacent-state-derived O_t^{cont} supports descriptive reconstruction but not an equivalent predictive claim. The predicted-coordinate bilinear branches in transition-coordinate-7F terminated with implementation errors and are excluded rather than imputed. See Source Data Fig. 12 and Source Data manifest.

Observed transition coordinates improved closure, but they used information from the transition being reconstructed. A pre-transition estimator recovered only part of that information. The best estimated-coordinate model reached macro $R^2 = 0.833$, retaining 6.7% of the observed-coordinate gain over the state-only baseline (Fig. 12d). History and context helped estimate O_l^{cont} , although macro R^2 remained between 0.41 and 0.61 in the tested variants (Fig. 12e). We therefore use O_l^{cont} as a descriptive coordinate unless it is estimated before the target transition. This left open whether the ordered states followed a simpler global route.

Supplementary Figure S9 | Transition-closure target-specific errors



transition-coordinate target-specific errors. **a**, Held-out model-by-target R^2 matrix. **b**, Median RMSE and MAE across targets; values are metric specific and not directly comparable across metric types. **c**, Highest and lowest model-ladder macro- R^2 values; the complete ladder is retained in Source Data. Observed-transition O_cont is descriptive unless estimated without target-layer information. See Source Data Fig. 12 and the Source Data manifest.

Are ordered layer-state trajectories close to an ideal shortest route?

Compact transition coordinates left open whether the observed route through layer-state space had a simple geometric form. We proposed the strict conjecture that ordered trajectories approximated an ideal geodesic relative to the chord joining the first and last analysed states. The order-shuffle control retained the same states and endpoints while permuting only the interior layer order; alignment, curvature and detour were then recomputed. The conjecture had two levels of prediction: real layer order should be more aligned and less curved or detoured than these shuffles, and an exact account should leave little residual motion. Real trajectories satisfied the first level: they were more aligned with the reference direction than shuffles, although the gain was heterogeneous across checkpoints (Fig. 13b), and curvature and detour were lower across the principal window and K comparisons (Fig. 13c,d and Supplementary Fig. S10).

Real trajectories show geometric bias without following exact geodesics

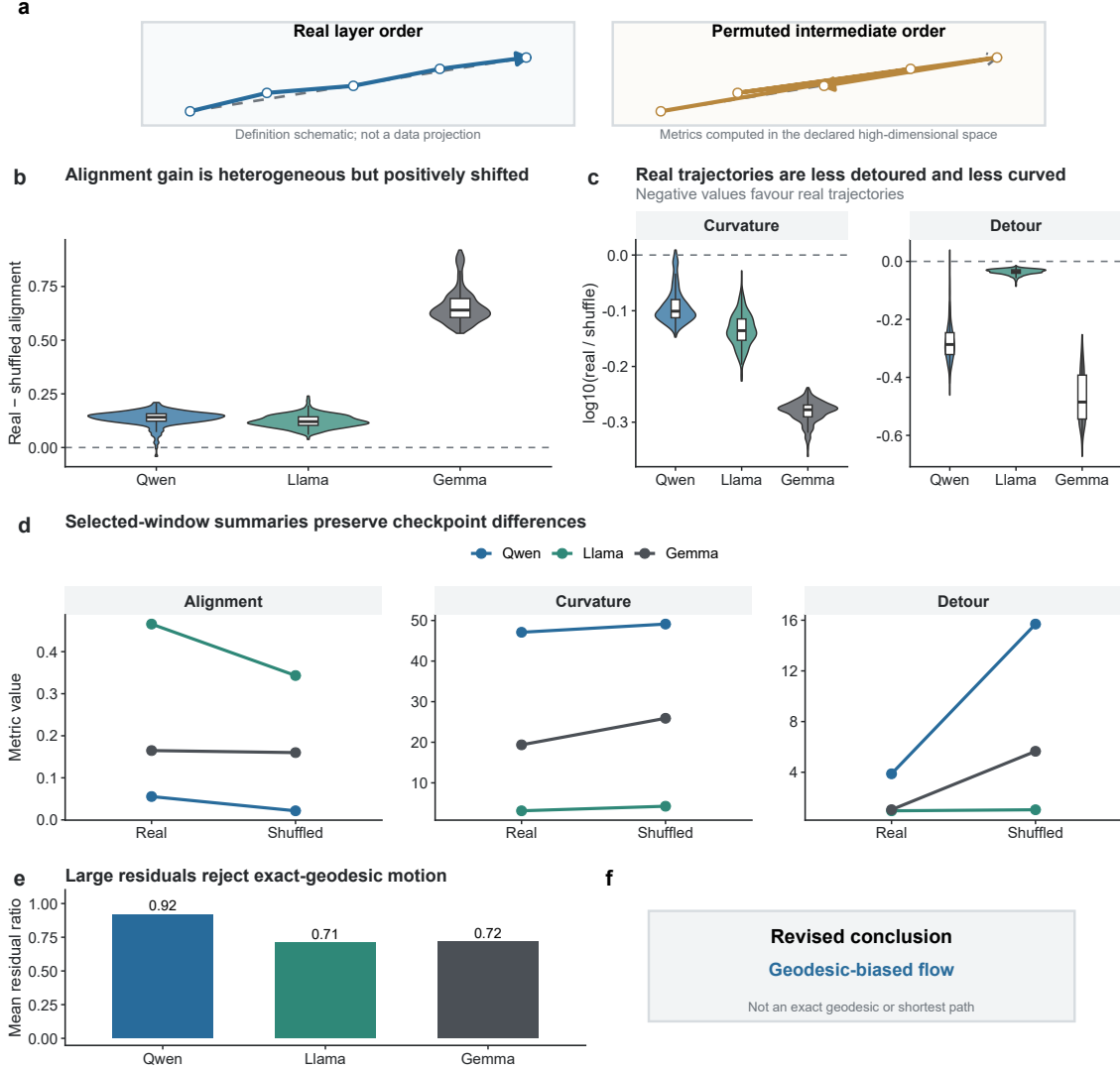
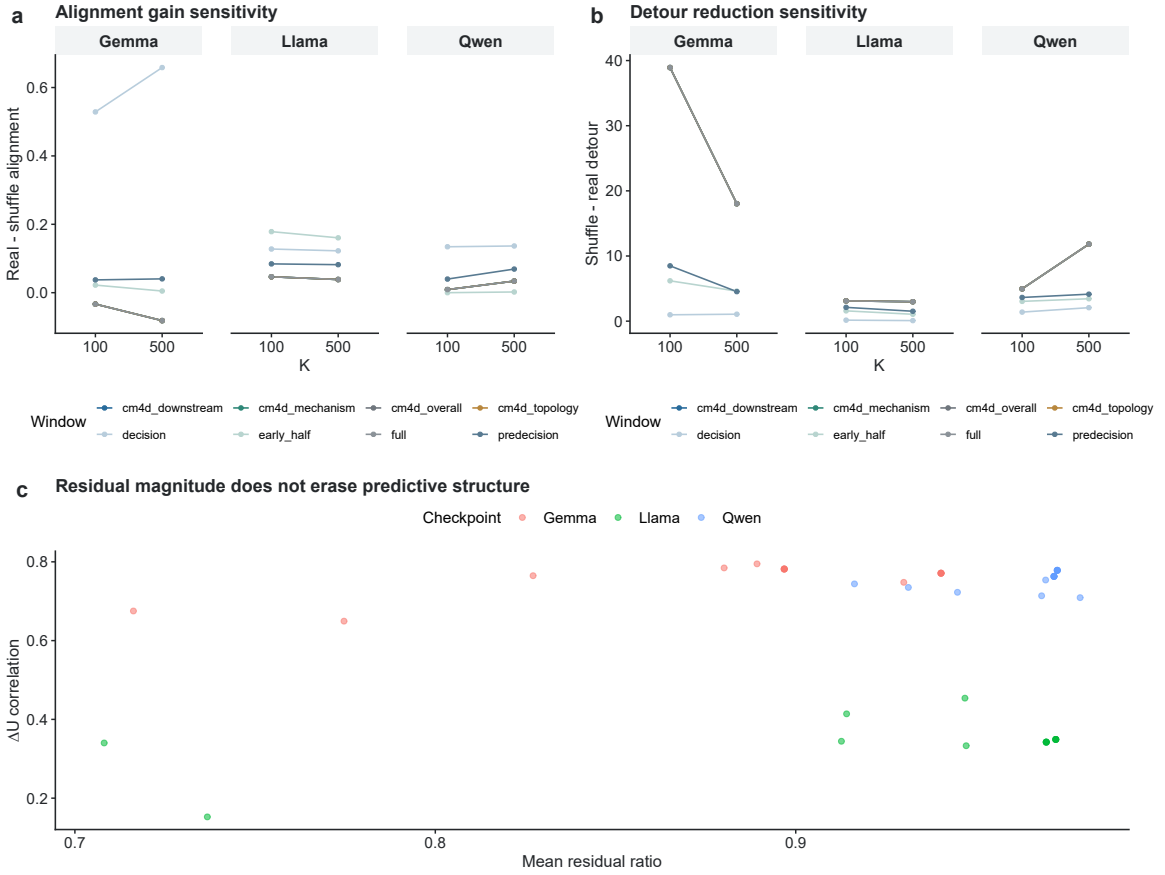


Fig. 13: Real trajectories show geometric bias without following exact geodesics. **a**, Control schematic comparing real layer order with an interior-layer permutation relative to a declared endpoint direction. The panel explains the test; real-data trajectory projections appear in Fig. 11b. **b**, Decision-window alignment gain for 1,296 checkpoint-prompt observations at $K = 500$, defined as real minus shuffled mean alignment. **c**, Detour and curvature contrasts expressed as $\log_{10}(\text{real}/\text{shuffle})$; negative values favour real layer order. **d**, Real and shuffled values in the separately selected best-bias window for each checkpoint. **e**, Mean real-trajectory residual ratio. This norm ratio is not a percentage of explained variance. Its large value excludes exact-geodesic motion, whereas geometric bias is established by the paired contrasts in panels b-d. **f**, The retained evidence supports geodesic-biased flow relative to the declared order shuffles, not exact geodesics. See Source Data Fig. 13 and Source Data manifest.

Supplementary Figure S10 | Geometry window, K and reference sensitivity



Geometry window, K and reference sensitivity. **a**, Real-minus-shuffle alignment by window and K. **b**, Shuffle-minus-real detour by window and K. **c**, Residual ratio against ΔU correlation. The retained claim is geodesic-biased flow, not exact geodesic following. See Source Data Fig. 13 and the Source Data manifest.

Exact-geodesic dynamics were decisively excluded. Mean residual ratios were 0.92 for Qwen, 0.71 for Llama and 0.72 for Gemma (Fig. 13e), leaving substantial motion outside the reference direction. These norm ratios are not percentages of explained variance and do not measure the significance of the aligned component. Geometric bias is established by the paired real-versus-order-shuffle contrasts in alignment, detour and curvature. Real trajectories were consistently more organized under those comparisons despite their large residuals. We call this combination geodesic-biased flow: directional organization without exact geodesic following. The residual then became a second empirical object rather than an error term to discard.

Are deviations from the reference direction signal or noise?

Once exact-geodesic motion was rejected, the unexplained component admitted two competing interpretations: nuisance deviation around an organized direction, or a second source of structured information. Each observed transition was summarized by a reference-aligned proxy and an off-direction residual proxy; the two feature families were evaluated separately and together under the same grouped splits. This is an implemented proxy decomposition, not an exhaustive orthogonal decomposition of hidden-state space. Under the noise-only account, residual features should add no stable information beyond the reference-aligned family. Residual-only features instead retained ΔU information across windows and checkpoints (Fig. 14b). Combined projection and residual features generally matched or exceeded either family alone, although effect size and the best feature family varied by checkpoint (Fig.

14b,c). The two proxy families were likewise complementary for ΔU and mechanism-label separation (Fig. 14d,e).

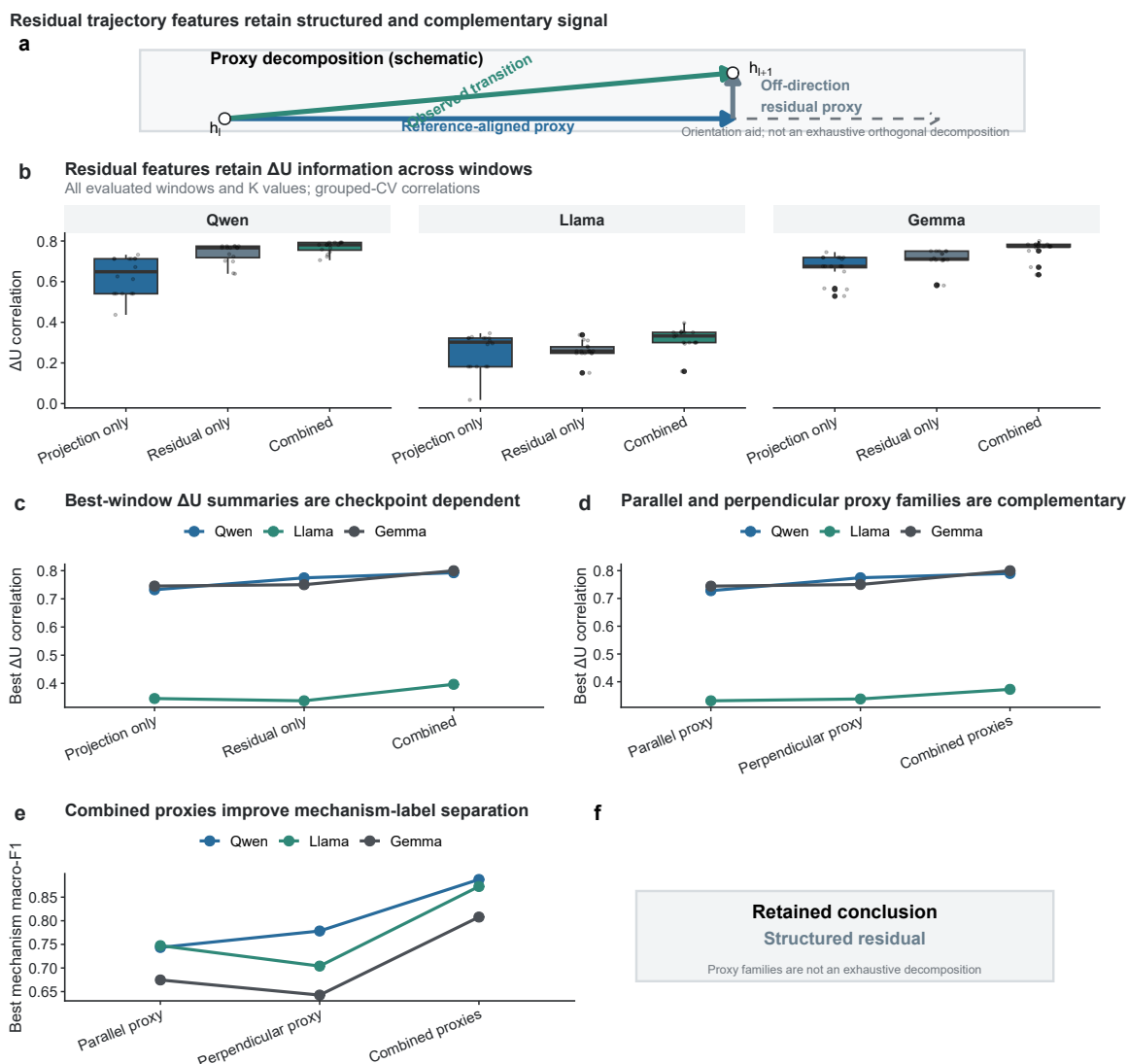


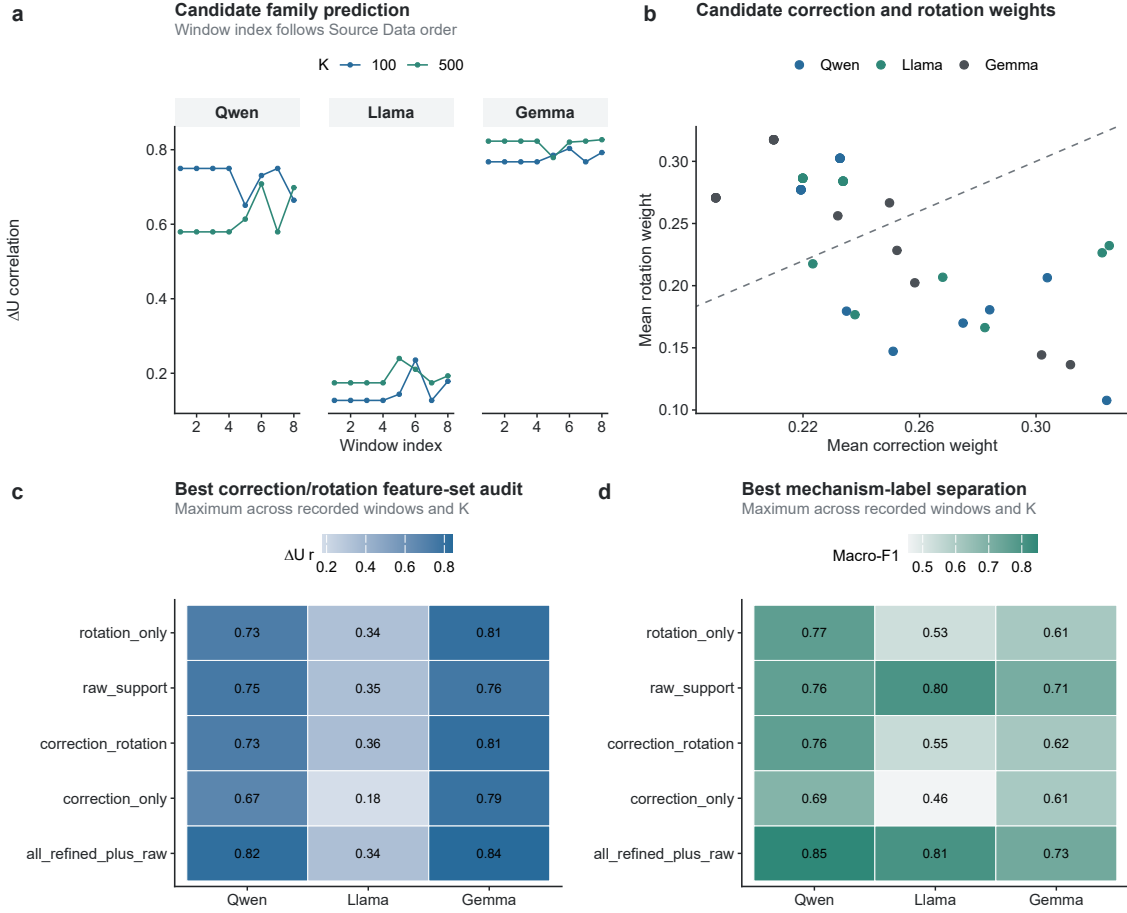
Fig. 14: Residual trajectory features retain structured and complementary signal. **a**, Definition schematic for the residual-feature analyses feature logic. An observed transition is represented by a reference-aligned proxy component and an off-direction residual proxy; the vector drawing is an orientation aid. Reference-aligned, residual and combined feature families are evaluated separately. These are implemented proxy families rather than an exhaustive orthogonal decomposition of the full hidden-state space. **b**, Grouped-cross-validation correlation with ΔU for projection-only, residual-only and combined feature sets over all declared window labels and K values. Residual-only signal is present across checkpoints, but Llama provides a counterexample to any claim that residual features always outperform projection features. **c**, Best-window ΔU correlations from projection-residual feature audit. Combined features are strongest in each checkpoint summary, with substantial checkpoint heterogeneity. **d**, Best ΔU correlations for parallel, perpendicular and combined proxy families in parallel-perpendicular proxy audit. **e**, Mechanism-label macro-F1 for the same proxy families. Combining the two families improves the best summary in all three checkpoints. **f**, Retained conclusion. Residual features are structured rather than disposable noise, while their stable taxonomy remains open. See Source Data Fig. 14 and Source Data manifest.

These results reject the noise-only account of off-direction motion. Residual features carried stable information beyond the reference-aligned family, and the combined representation was generally strongest. Exploratory correction-and-rotation tests found more consistent correction-related measures than rotation measures (Supplementary Fig. S11). The present decomposition does not exhaust the residual, but it

establishes that the non-geodesic component is structured. The final experiment asked whether the retained coordinates were also operationally discriminating.

Supplementary Figure S11 | Exploratory residual-family taxonomy

Exploratory only: correction is more consistent; rotation is not a retained stable independent family



Exploratory residual-family taxonomy. **a**, Candidate family prediction across windows and K. **b**, Candidate correction and rotation weights. **c,d**, Best ΔU correlation and mechanism macro-F1 across recorded windows and K for correction/rotation feature sets. This taxonomy is exploratory: correction is more consistent, whereas rotation is not retained as a stable independent family. See Source Data Supplementary Fig. S11 and the Source Data manifest.

4 Testing a bounded operational consequence

Can the revised coordinates support specific internal modulation?

The preceding analyses were passive measurements. We next asked a narrower operational question: could the retained coordinates select actions that changed the internal endpoint more specifically than fixed and matched-shuffle alternatives? Mechanism and policy models were fitted on training relation graphs, used to select actions for held-out graphs, and compared with a fixed action and three label-shuffle null families. No held-out graph contributed to coordinate fitting, action-label construction or classifier training. The endpoint, specificity, was defined as the target absolute ΔU shift minus the non-target absolute shift. Classifier accuracy and macro-F1 values are training diagnostics, whereas specificity was evaluated on held-out graphs.

Mechanism-aware policies exceeded fixed action combinations in all three representative checkpoints

(Fig. 15b), but the raw specificity gain was much smaller in Llama than in Gemma. The mechanism labels were assigned from the controlled prompt conditions before fitting; they were not discovered from generated answers or inferred model behaviour. Against the joint mechanism-and-policy shuffle, observed specificity yielded $z = 4.38$ for Qwen, 4.23 for Llama and 4.02 for Gemma, and exceeded the empirical 95th percentile in each checkpoint (Fig. 15c,d). Results varied across null families. Llama fell below the 95th percentile of the mechanism-policy-pairing shuffle considered alone. All checkpoints exceeded the policy-label and joint-shuffle controls (Supplementary Fig. S12a,b). Policy actions were also imbalanced, with the 0/0 action common in every checkpoint (Supplementary Fig. S12c).

Mechanism-aware policies modulate internal dominance under matched nulls

Internal modulation endpoint only; no answer-correctness or deployment-control conclusion

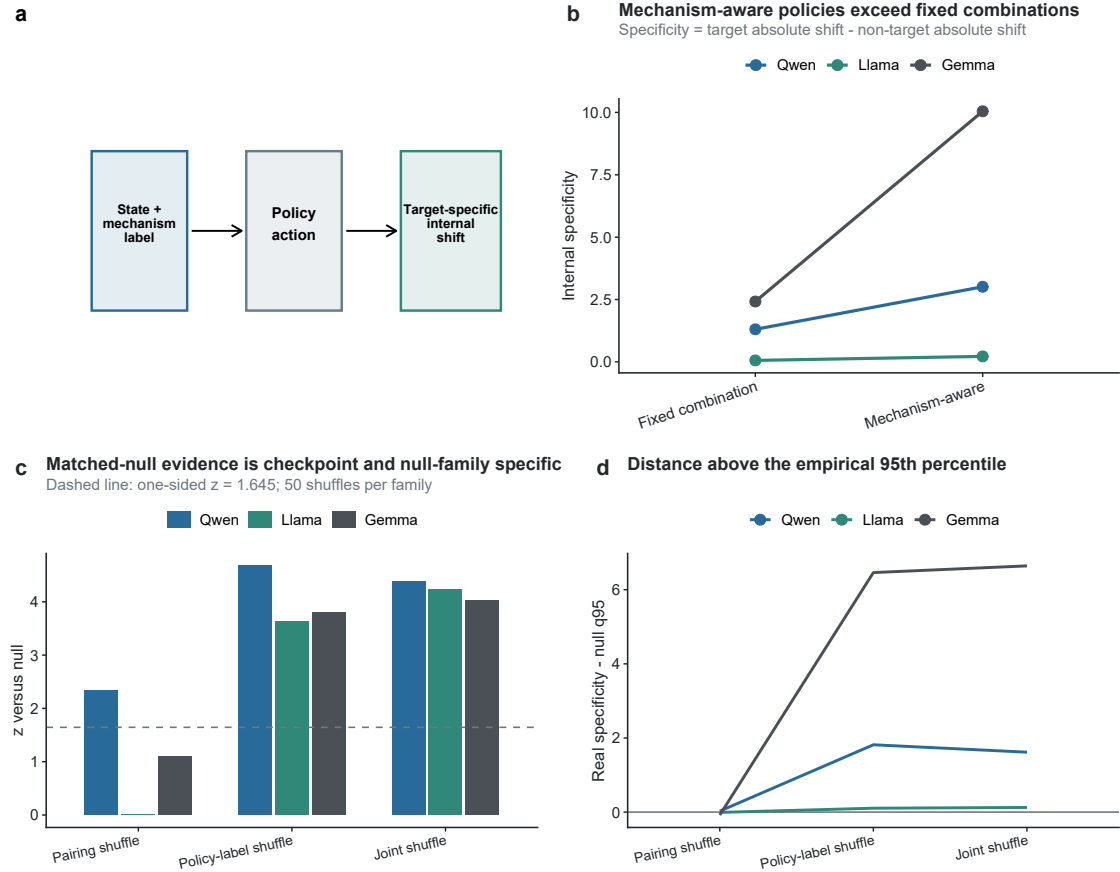
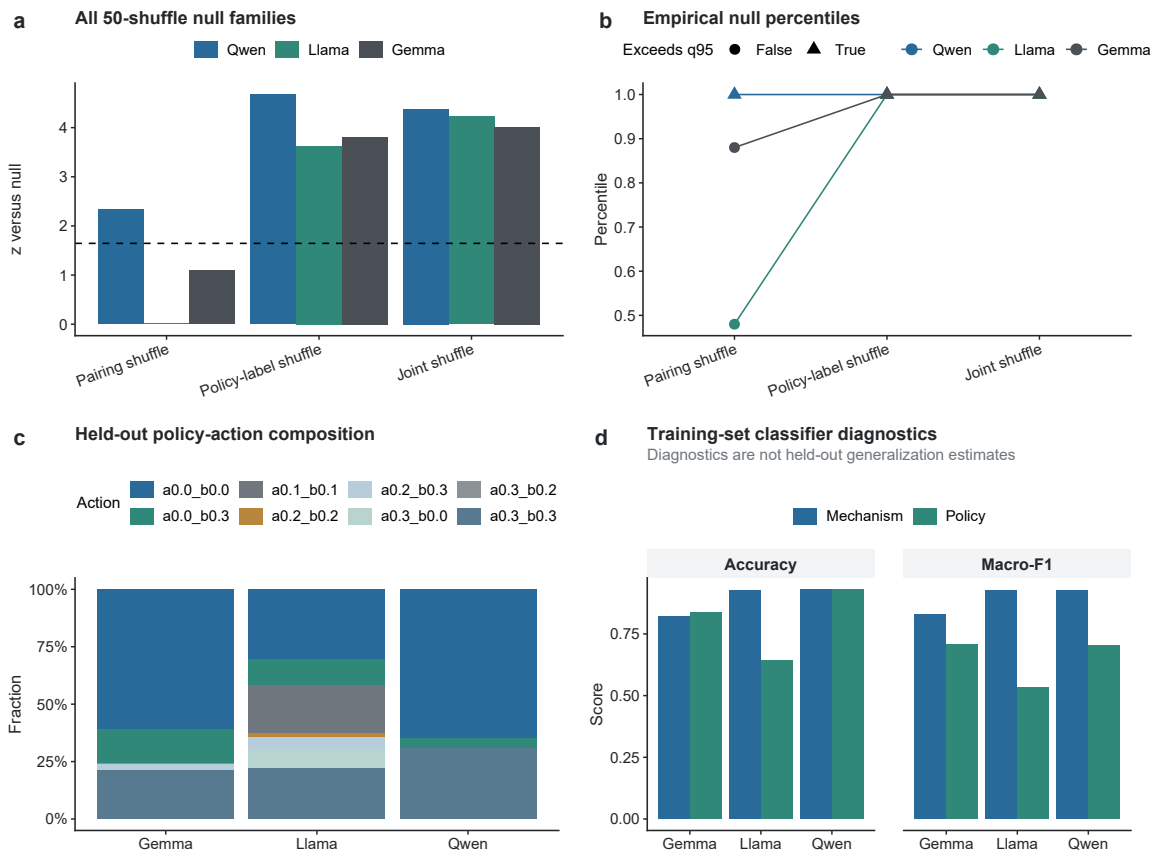


Fig. 15: Mechanism-aware policies modulate internal dominance under matched nulls. **a**, Audit workflow: checkpoint state and mechanism label determine an internal policy action, evaluated by a target-specific internal-shift endpoint. **b**, Internal specificity (closure-condition absolute shift minus non-closure absolute shift) for fixed-combination and mechanism-aware policies. Lines join results within each representative checkpoint. In the action labels, none denotes 0/0 operator/order-precursor weights. **c**, Standardized separation of the observed mechanism-aware specificity from three 50-shuffle null families. The dashed line marks a one-sided normal-reference threshold of 1.645 and is shown only as a visual guide; empirical quantiles are reported in panel d. **d**, Difference between observed specificity and the empirical 95th percentile of each matched null. Qwen, Llama and Gemma denote representative checkpoints from the respective families. Mechanism and policy classifier values used in this experiment are training diagnostics, not held-out generalization estimates. The endpoint establishes internal modulation specificity only; it does not establish answer correctness, behavioural benefit or deployment control. See Source Data Fig. 15 and the Source Data manifest.

Supplementary Figure S12 | ASA complete audit package
Internal specificity only; null-family heterogeneity and class imbalance are retained



internal-modulation complete audit package. **a,b**, Standardized and empirical-percentile results for all three 50-shuffle null families. **c**, Held-out policy-action composition; $a_{0.0_b0.0}$ denotes 0/0 operator/order-precursor weights. **d**, Training-set mechanism and policy classifier diagnostics, which are not held-out generalization estimates. Null-family heterogeneity and action imbalance are retained. The endpoint concerns internal modulation only, not answer correctness or deployment control. See Source Data Fig. 15, Source Data Supplementary Fig. S12 and the Source Data manifest.

These experiments close a controlled observation-to-operation loop. Trajectory-informed policies produced target-specific internal changes that fixed and label-shuffled alternatives did not reproduce. The result establishes operational relevance for the retained coordinate within the internal dominance endpoint. Whether this modulation changes answer selection or correctness is a separate behavioural question that was not tested here.

5 Task, prediction, initialization and output-boundary controls

Does layer-order geometry recur beyond binary relation graphs?

We tested a 64-problem lexical-category task, a 64-problem addition task and a deterministic 96-item ARC-Challenge validation subset presented with all four original choices. Across all nine task-checkpoint combinations, real-order alignment exceeded every one of 50 endpoint-preserving shuffles; null-standardized effects ranged from 3.21 to 7.84 (Fig. 16a,b). Task-specific one-dimensional coordinates were fitted on 44 controlled problems and evaluated on 20 held-out problems. They exceeded the training-label-shuffle 95th percentile in five of six task-checkpoint tests. The numerical relation-graph ΔU axis was not transferred. Llama accuracy was 0.333 on ARC and 0.315 on addition, showing that chord-directed geometry can persist when task behaviour is weak.

Can an ordered partial trajectory forecast a later candidate boundary?

Ridge models forecast the exact final clean-minus-conflict output-logit margin from the current and earlier decision-window margins, using 67 relation graphs for fitting and 29 held out. The order null retained the current value but independently permuted all earlier values within each prompt before refitting. At the earliest cutoff above the 95th percentile of 50 nulls, held-out R^2 was 0.574, 0.393 and 0.714 and boundary AUROC was 0.857, 0.837 and 0.984 for Qwen, Llama and Gemma (Fig. 16c). This is a prospective reduced-order forecast of one declared candidate contrast, not an autonomous state equation or a task-general predictive theory.

Is chord-directed order specific to trained parameters?

We instantiated three full random models from each Qwen, Llama and Gemma configuration. Mean random-initialization alignment gains were 0.0768, 0.1028 and 0.0879, versus 0.0317, 0.0403 and 0.0499 in the matched trained checkpoints (Fig. 16d). All nine random gains were positive and exceeded their trained counterpart. Chord-directed order is therefore not evidence of learned reasoning by itself; across these configurations it is compatible with an architectural scaffold on which training adds task-conditioned content.

Does internal displacement explain the original Qwen boundary failure?

Under the original action budget, candidate crossings occurred in 0 of 87 Qwen, 9 of 45 Llama and 6 of 56 Gemma closure prompts. Qwen nevertheless moved 86 of 87 prompts towards clean and shifted its mean margin from -5.03 to -2.54 ; the maximum remained -0.56 . Cross-fitted maps from ΔU displacement to margin displacement reached pooled $R^2 = 0.850$ and separated the 15 original-budget crossings with AUROC 0.976. Together with 94% saturation at the maximum original action, the Qwen result identified an exhausted actuator budget rather than absent output leverage.

Can a bounded actuator reach the declared emitted-token boundary across checkpoints?

We expanded the bounded action library independently in all three checkpoints while fitting coordinates, transition models and boundary policies on training relation graphs only. Under one predeclared confidence-and-entropy gate, full-vocabulary top-1 changed from conflict to clean in 17/87, 13/87 and 9/87 held-out closure prompts in Qwen, Llama and Gemma, with 0/203, 2/203 and 0/203 non-closure changes (Figs. 16e and 17e). Ungated crossings were 19/87, 13/87 and 9/87, with collateral rates of 3.45%, 1.48% and 0.49%. A naively shuffled policy produced 20–36% collateral changes and was not an attribution null.

Fifty safety-matched reassignments per checkpoint preserved the gate-open set and every condition-by-layer action multiset. Their mean strict-flip counts were 17.64, 14.18 and 7.90, and observed assignments did not exceed the matched-null 95th percentile ($P = 0.804, 0.922, 0.255$; Fig. 16f). The evidence therefore supports boundary reachability by the gate and bounded actuator family, not fine-grained trajectory-to-dose assignment. Clean denotes the unperturbed-chain candidate and is generally not the rule-consistent closure answer; this is emitted-token candidate control, not accuracy improvement.

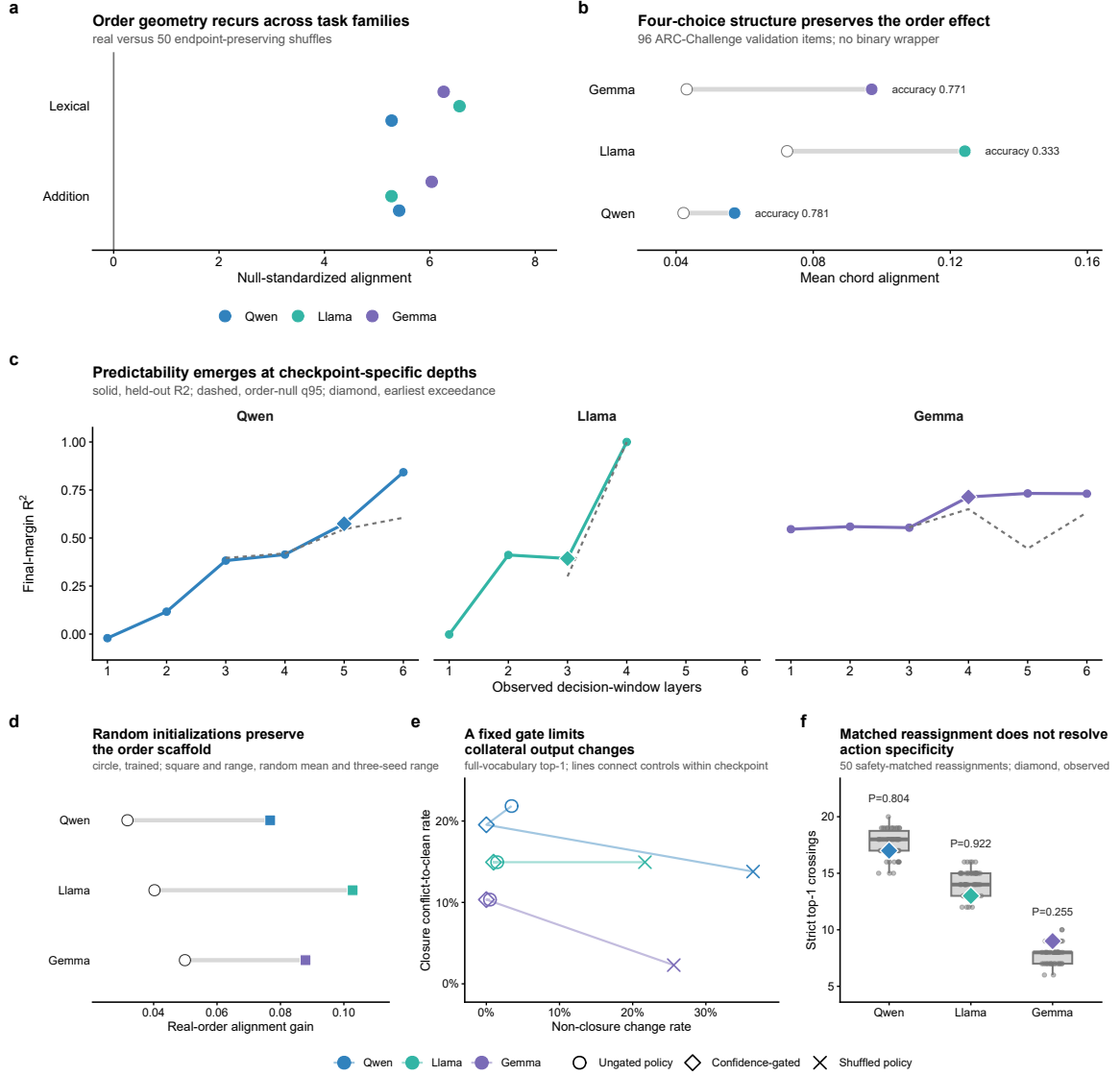
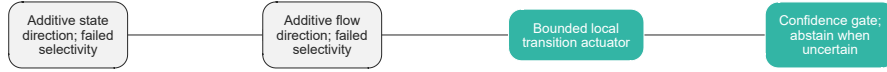


Fig. 16: Task, prediction, initialization and output-control boundaries. **a**, Null-standardized chord alignment in controlled lexical and addition tasks. **b**, Four-choice ARC-Challenge geometry without binary reduction. **c**, Held-out final candidate-margin R^2 for ordered partial histories and the 95th percentile of 50 order nulls. **d**, Trained checkpoints and three random initializations per architecture. **e**, Target crossings and non-closure changes for ungated, gated and shuffled-policy controls. **f**, Observed strict flips against 50 safety-matched action reassignments. Task coordinates remain task-specific; forecasting is candidate-defined; random-state geometry is not learned reasoning; candidate crossing is not correctness.

a Negative results changed the actuator object

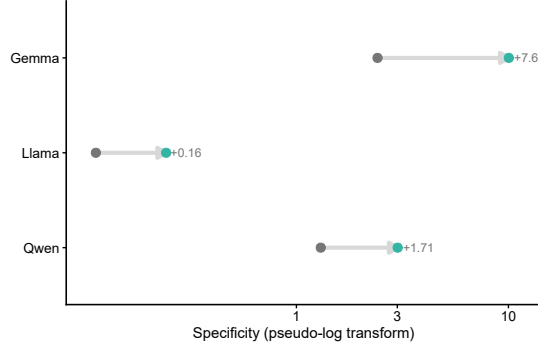
A correlated direction was not sufficient for selective internal control



The actuator moves internal trajectories and can cross a declared emitted-token boundary

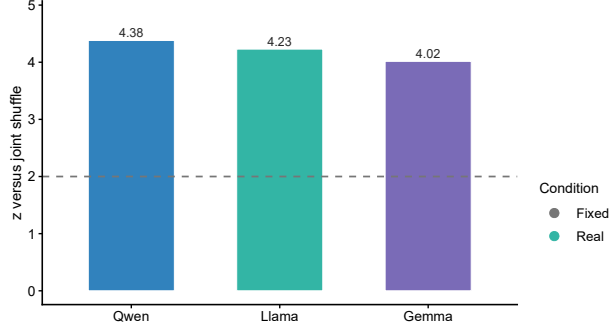
b Internal trajectory control is selective

held-out perturbation; output boundary tested separately



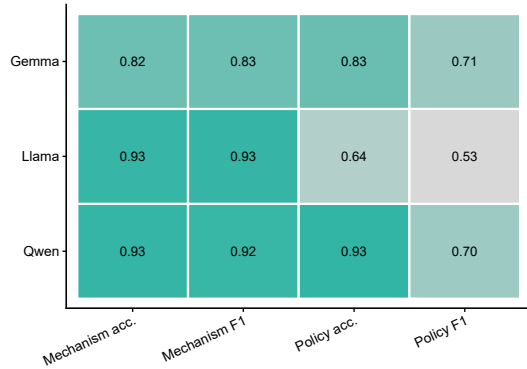
c All models exceed joint-shuffle nulls

dashed line: $z = 2$



d Classifier quality bounds interpretation

mechanism and policy models



e The declared output boundary is reachable across checkpoints

confidence-gated full-vocabulary top-1; target / non-target counts

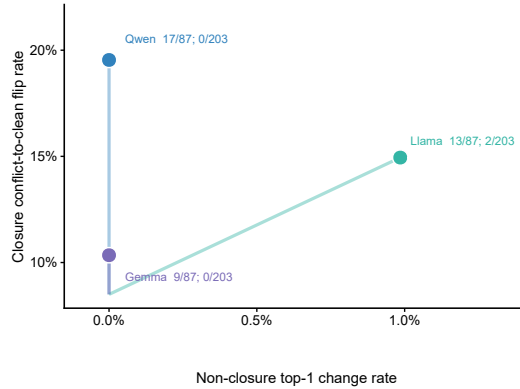


Fig. 17: A bounded local actuator controls trajectories and a declared output boundary. **a**, Additive failures, transition-actuator switch and confidence gate. **b**, Held-out internal specificity relative to fixed actions. **c**, Real policy relative to 50 joint-label shuffles. **d**, Mechanism and action-policy classifier diagnostics. **e**, Confidence-gated full-vocabulary conflict-to-clean top-1 changes and non-closure changes in Qwen, Llama and Gemma. Safety-matched reassignment is reported in Fig. 16f and does not support fine-grained action-assignment specificity.

6 Falsification-first separation of scaffold, execution and learning

Can random fixed maps preserve relational continuity?

To test whether cross-layer relational continuity required learned weights, we propagated normalized prompt states through random fixed maps while preserving architecture-specific dimensional schedules. The continuity criterion was tested across Qwen, Llama and Gemma configurations, lexical, addition and four-choice ARC prompts, and two independent seeds. All 18 architecture–task–seed conditions passed (Fig. 18a). This supports an architecture-compatible relation-propagation scaffold; it does not imply semantic or task-functional equivalence to a trained model.

Can preserved geometry substitute for native execution?

It could not. In Qwen, reversed and fixed-permutation executions retained positive chord contrasts but had zero full-vocabulary top-1 agreement with native execution, candidate agreement of 0.479 and mean Jensen–Shannon divergence of 0.692 (Fig. 18b). Thus a state set can retain finite-path regularity while implementing a different computation. The functional contribution lies in the ordered execution, not in alignment alone.

Does trajectory history predict beyond the current state?

A cross-task incremental test compared current-state predictors with predictors augmented by ordered-prefix summaries. Six of 18 initial conditions passed, all in addition; lexical prompts passed 0/9. An independent mixed-arithmetic replication also passed 0/9 (Fig. 18c). The earlier relation-graph forecast therefore remains a bounded candidate-margin result. The tested histories do not establish a general path-memory law, and the current high-dimensional state may already summarize much of the usable history.

Is chord alignment a direction-only property?

A Llama metric audit separated direction and radial scale. Direction-only trajectories passed 0/7 task gates, whereas norm-only fixed-direction trajectories passed 7/7. Applying a frozen radial exponent produced monotonic metric changes in all seven tasks (Fig. 18d). Chord alignment is consequently sensitive to radial structure under the declared normalization and cannot by itself identify a directional mechanism.

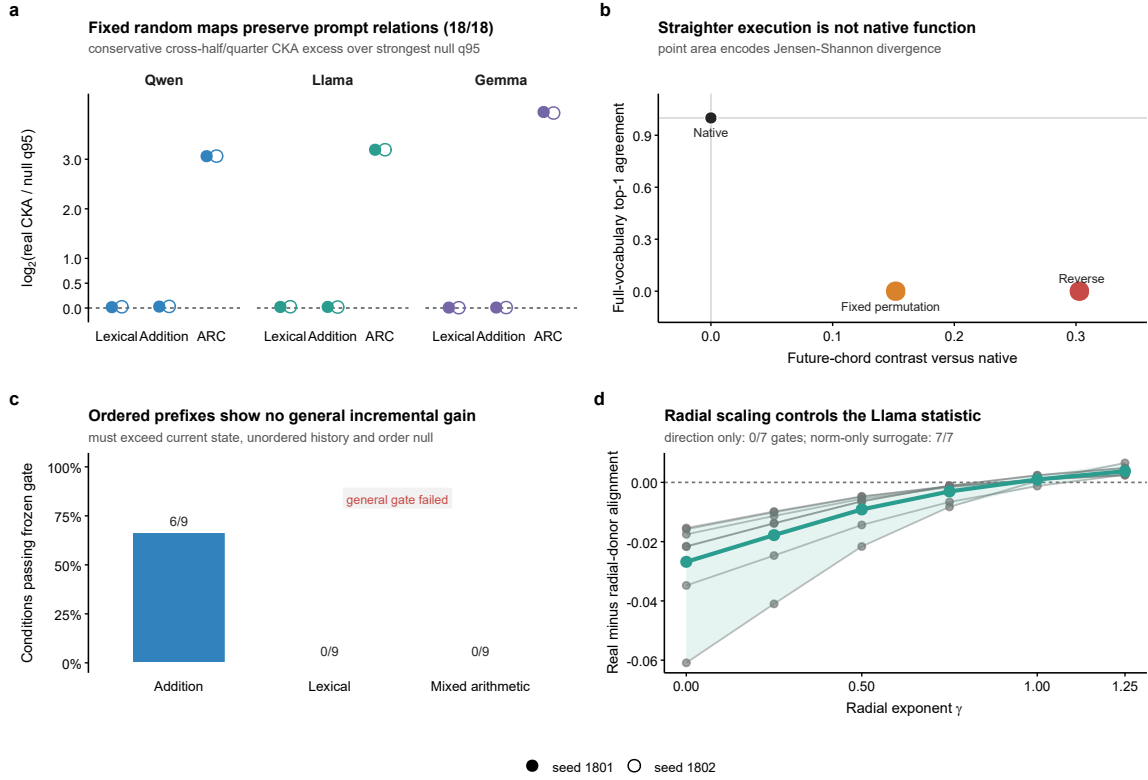


Fig. 18: Falsification controls bound the interpretation of layer-order geometry. **a**, Random fixed-map relational-continuity contrasts across three architecture configurations, three task families and two seeds; all 18 declared conditions pass the positive-contrast gate. **b**, Reversed and fixed-permutation Qwen executions retain positive geometry but lose native full-vocabulary outputs. **c**, Ordered-prefix summaries add held-out signal beyond current state only in 6/9 addition conditions and in neither lexical nor mixed-arithmetic tests. **d**, Llama chord metrics vary monotonically under frozen radial rescaling. These controls separate architecture-compatible propagation, native ordered execution and metric sensitivity.

What does training add to the architectural scaffold?

We swapped embeddings (E), transformer blocks (B), final normalization (N) and output head (W) between initialized and trained Pythia-160m checkpoints in all 16 component combinations. On 53 held-out lexical cloze prompts, accuracy rose from 0.472 at initialization to 0.849 when all components were trained. No isolated trained component exceeded 0.604, and the superadditive margin excess was 2.621 (95% confidence interval, 1.252–3.987; Fig. 19a,b). Function therefore depended on compatibility among input, transition and output components rather than on one trained component in isolation.

Across seven training checkpoints, model-native candidate-coordinate order gain was absent at step 0, appeared from step 1,000 and correlated with accuracy (Spearman $\rho = 0.75$; Fig. 19c). A coordinate fitted at the final checkpoint transferred with macro F1 0.041 at steps 0 and 128, 0.701 at step 1,000 and 1.0 by step 16,000 (Fig. 19d). By contrast, a coordinate refitted within each checkpoint separated lexical categories perfectly even at random initialization, so that control was rejected as evidence of learning. These results separate architecture-supported relational propagation from training-organized task coordinates and component compatibility.

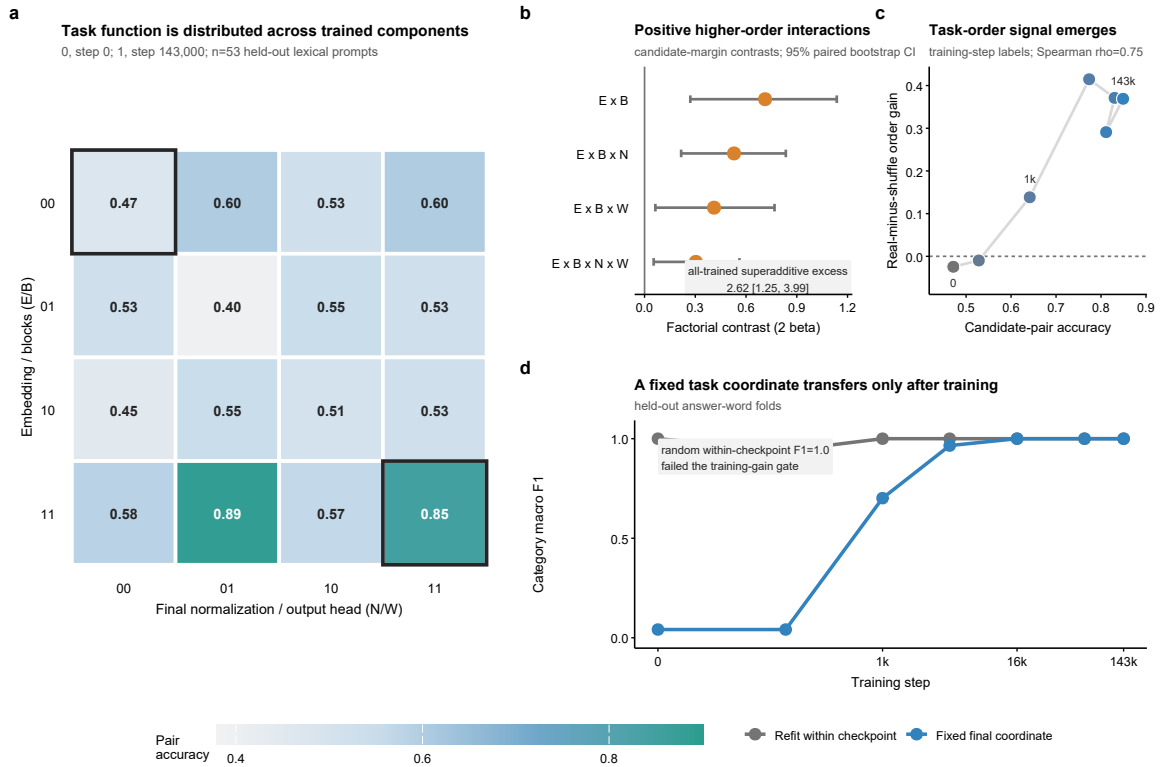


Fig. 19: Training organizes compatible components and task coordinates. **a**, Accuracy for all initialized/trained combinations of embeddings, blocks, final normalization and output head. **b**, Higher-order interaction terms; the fully trained system has a positive superadditive margin beyond additive component gains. **c**, Model-native candidate-coordinate order signal emerges during Pythia training and tracks held-out accuracy. **d**, A coordinate fixed at the final checkpoint transfers poorly at initialization and becomes predictive during training, whereas within-checkpoint category separation is already perfect at initialization and is therefore a negative control rather than evidence of learning.

7 Discussion

Across the tested checkpoints, realized transitions formed a measurable internal process and local perturbations crossed a declared emitted-token boundary. The new controls separate three levels that raw geometry had conflated: architecture supports relational propagation, training organizes compatible task coordinates and native layer order executes functionally consequential transitions. The ordered trajectory supplies the common empirical object on which these levels can be compared and perturbed. This is process-level causal evidence with bounded behavioural leverage, not a universal dynamical law.

The measurements draw on restricted mathematical and physical ideas. Depth indexes a discrete trajectory; checkpoint comparison resembles functional registration; ΔU adopts reaction-coordinate compression without claiming diffusion optimality, a committor or a slow variable. Transition closure tests compact representation of an observed update, whereas chord alignment, detour and turning angle test finite-path directional efficiency. The actuator separates state estimation from local action, and its gate abstains under uncertainty. These are sources for executable tests, not claims of physical identity.

The cross-disciplinary implication is a transferable audit, not a mechanism analogy. Given measurable states and transitions, ask whether a process object supports a compressive coordinate, rejects an endpoint-only explanation under an order-sensitive null, and responds at a predeclared boundary to targeted perturbation. Artificial systems make these tests unusually direct. A systems-neuroscience analogue would compare population trajectories with neuron-wise or static summaries, then perturb an inferred transition against a behavioural boundary. Neural trajectories evolve in physical time, whereas transformer depth is a computational index; appropriate states, timescales and interventions would differ. No shared dynamical equation is implied.

This view complements component-level interpretability. Circuits can identify which parts realize a local computation; probes and sparse features can identify information available at a state; readbacks can expose output-facing tendencies; and causal interventions can localize consequential variables. A trajectory-level object asks a different question: how do those local objects participate in an ordered process? The answer can then guide where component analyses and interventions should be placed.

Rival explanations remain. Proxy ranking is task- and classifier-dependent. Geometry recurred in lexical, addition and ARC tasks, but task coordinates transferred less consistently and the relation-graph ΔU axis was not reused. Functional windows came from one non-nested exploratory scan. The relation-graph margin forecast did not generalize to an incremental path-memory law; pre-transition estimates also recover little of the descriptive transition-coordinate gain. The training audit used one small Pythia checkpoint series and one lexical task, and component swaps are artificial states rather than natural training trajectories. Stronger random-state geometry, radial sensitivity and reordered executions jointly show that geometry alone is neither learned reasoning nor functional fidelity. Gated control crossed 17, 13 and 9 of 87 closure prompts, but changed 2 of 203 non-closure outputs in Llama, and matched reassignment supported no fine-grained dose specificity. These are not safety guarantees, candidate crossing is not correctness, and intervention remains within one relation-graph family.

A broader LLM-dynamics landscape may include architecture supplying a propagation scaffold, training organizing a task-compatible representational substrate, prompts conditioning initial states and available directions, and transformer layers realizing local transitions. The Pythia audit provides a bounded training-time bridge: fixed task coordinates emerge and function is distributed across compatible components. It does not map a complete training-shaped semantic geometry or establish how prompt-conditioned directions are formed. Those questions require new experiments rather than extrapolation from the present trajectory data.

Timestamped records show that the core mapping, geometry and intervention sequence was developed through a human-directed, AI-assisted hypothesis-to-audit loop, with later audits revising interpretations.

This single case lacks a no-AI counterfactual and records achieved throughput rather than an AI-caused productivity gain. It nevertheless shows how AI can extend retrieval, coding, batch planning and consistency checks while object selection, falsification, interpretation and evidentiary authority remain human.

The transferable lesson is to choose the empirical research object before its coordinates. The broader question is when a chosen object makes distributed computation observable, computable and controllable. We establish one bounded artificial case; extending it towards a general theory of cognition requires concepts and experiments beyond this paper.

8 Methods

Purpose and scope

These Methods describe the checkpoints, prompt sets, extraction procedures, controls and analyses used in *High-dimensional state trajectories reveal transformer inference dynamics*. Public-facing workflow names are descriptive; file-level provenance and hashes are retained in the Source Data manifest.

The direct empirical scope is inference-stage trajectory dynamics across Qwen, Llama and Gemma. Training-shaped semantic structure, prompt-conditioned initial conditions and local transformer unfolding are discussed as the broader landscape, but the present experiments directly test the inference-stage layer-state trajectory process.

Model checkpoints

Analyses used representative instruction-tuned checkpoints from the Qwen, Llama and Gemma families. Model weights are not redistributed and remain subject to provider access conditions.

Model family	Checkpoint	Revision/SHA	Licence tag	Context/config note
Qwen	Qwen/Qwen2.5-1.5B-Instruct	989aa7980e4cf806f80c7fef2b1adb7bc71aa306	apache-2.0	Public config reports max_position_embeddings=32768; tokenizer config reports model_max_length=131072.
Llama	meta-llama/Llama-3.2-1B-Instruct	9213176726f574b556790deb65791e0c5aa438b6	llama3.2	Hugging Face metadata is public but files are manually gated; local config reports max_position_embeddings=131072.
Gemma	google/gemma-2-2b-it	299a8560bedf22ed1c72a8a11e7dce4a7f9f51f8	gemma	Hugging Face metadata is public but files are manually gated; local config reports max_position_embeddings=8192 and sliding_window=4096.

Checkpoint metadata was checked on 21 June 2026. The machine-readable records are indexed in the Source Data manifest.

Prompt inventory

The controlled prompt inventory contains 96 prompts per checkpoint across 12 conditions. It comprises eight complete relation-graph blocks, each containing every condition once. We used the first eight complete blocks in the frozen source order, rather than sampling individual rows, to preserve condition

balance and graph-grouped splitting while holding the extraction and null-control budget constant across checkpoints. This fixed-prefix design was not randomized; conclusions therefore concern the controlled subset and are checked across checkpoints rather than treated as population estimates. Prompt-level metadata and condition summaries are indexed in the Source Data manifest. Prompt conditions are experimental labels, not general natural-language task categories.

Hidden-state trajectory extraction

For the controlled proxy-audit artifacts reused by the projection-selection geometry control, hidden-state tensors were extracted from Hugging Face model outputs with hidden-state return enabled and key-value caching disabled. For each batch item and transformer layer index l , the stored vector was:

$$H_{m,p,l} = H^{(l+1)}[b, t_{\text{last}}, :].$$

The $l + 1$ indexing excludes the embedding hidden state. The token position was the last non-padding token from the attention mask, with tokenizer padding side set to left. Models were loaded as float16 on CUDA or float32 on CPU; saved hidden-state arrays were cast to float32. Extraction used a maximum input length of 320 tokens, batch size 4 and random seed 42, without applying a chat template.

Model family	Hidden-state array shape	Analysed layers
Qwen	[960, 28, 1536]	7-25
Llama	[960, 16, 2048]	4-14
Gemma	[960, 26, 2304]	6-23

The projection-selection geometry control used this 96-prompt controlled subset from each 960-row artifact. The machine-readable extraction table is documented in the Source Data manifest.

Projection-selection geometry controls

Projection-selection geometry controls compared real output weights with row-permuted weights and Gaussian projections. Row-permutation controls permuted vocabulary rows. Gaussian controls matched the analysed real projection matrix by per-dimension mean and standard deviation, then rescaled row norms to the real row-norm distribution.

The projection-selection geometry control used five repeats for the row-permuted and Gaussian projection conditions, with random seed 42. Projection preservation was computed as Pearson and Spearman correlation between pairwise cosine-distance vectors in hidden-state space and readback-centre space.

Readbacks are diagnostic observables. These controls do not imply that vocabulary-facing coordinates exhaust the mechanism coordinate system.

Strict-predecision matched-capacity object audit

Strict-predecision matched-capacity audit used the frozen 960-observation, 80-relation-graph dataset for each checkpoint. Both hidden and vocabulary-readback proxies were restricted to layers before the declared decision window. The vocabulary proxy comprised adjacent TopK-centre shifts over the predecision layers plus their mean, sum, maximum and standard deviation. The hidden proxy was the mean hidden state over the same predecision window. Neither proxy included decision-window states, R_l , dR_l , candidate logits or generated final labels.

We evaluated matched retained dimensions $d \in \{2, 4, 6, 8\}$. Within each of five relation-graph-grouped training folds, missing values were median-imputed, features standardized and PCA fitted with exactly d components; the fitted transforms were then applied to held-out relation graphs. Ridge regression with $\alpha = 10$ predicted ΔU , and class-weighted logistic regression predicted the controlled mechanism labels. Reported R^2 , Pearson correlation and macro-F1 were calculated from pooled out-of-fold predictions. Intervals in Supplementary Fig. S13 are percentile 95% intervals from 500 bootstrap resamples of relation-graph groups. The audit tests temporal proximity and retained dimensional capacity together. It does not make the controlled mechanism labels independent of the prompt-condition design.

Parameter-selection status

Ridge regularization $\alpha = 10$ was fixed across checkpoints, proxy families and folds rather than selected separately to maximize a reported score. It provides common moderate shrinkage after feature standardization; the present analysis does not claim that 10 is an independently optimized value. The principal geometry analyses used $K = 500$ as the declared high-resolution neighbourhood setting, while the associated K scans and robustness panels report the available lower-K comparisons. The fractional decision interval $0.70 \leq l/L \leq 0.92$ was fixed before fitting ΔU to isolate the late decision region identified by the preceding layer-flow analyses; it was not selected by optimizing ΔU prediction. By contrast, functional-window and best-bias summaries were selected from complete scans on the same controlled dataset and are explicitly treated as exploratory rather than independently confirmed optima.

Setting	Value	Selection status
Common random seed	42	Fixed implementation seed; shuffle replications used declared offsets.
Ridge regularization	$\alpha = 10$	Fixed after standardization; not tuned per checkpoint or proxy.
Principal neighbourhood size	$K = 500$	Declared high-resolution setting; available K sensitivity is reported separately.
Matched retained dimensions	2, 4, 6, 8	Prespecified capacity audit grid.
Main grouped cross-validation	5 folds	Relation graph was the split unit.
Decision interval	0.70-0.92 relative depth	Fixed before ΔU fitting from the preceding late-transition analysis.
Matched-capacity uncertainty	500 graph bootstraps	Applied to pooled out-of-fold predictions.
Internal-modulation split	30% held-out graphs	Fixed graph-grouped test partition.
Intervention null replication	50 shuffles per family	Pairing, policy-label and joint null families.

Control families and the explanations they test

Control	Preserved	Disrupted	Question tested
Row-permuted output projection	Output-row set and projection scale	Learned token-row identity	Does learned vocabulary identity supply the preserved geometry?
Matched Gaussian projection	Projection-selection pipeline and output dimension	Learned output projection	Is preservation specific to the learned projection?
Random-K selection	Number of selected rows	Conditional TopK selection	Does conditional selection contribute to preservation?
Predecision exclusion	Graph and condition structure	Decision-window features	Does association survive direct target-window removal?
Interior-layer order shuffle	States and trajectory endpoints	Natural interior layer order	Does ordered depth contribute geometric organization?
Mechanism/policy label shuffles	Action grid and endpoint calculation	Mechanism-action matching	Is internal specificity reproduced by mismatched policies?

ΔU and candidate-margin advantage-flow analyses

Broad-window and predecision advantage-flow analyses used a clean/conflict label readback margin:

$$R_l = \text{logit}_l(y_{\text{lead}}) - \text{logit}_l(y_{\text{comp}}).$$

For each graph, the clean-anchor trajectory was subtracted:

$$DR_l^{(g,c)} = R_l^{(g,c)} - R_l^{(g,c_{\text{anchor}})}.$$

Decision layers occupied the declared fractional interval $0.70 \leq l/L \leq 0.92$:

$$\mathcal{L}_{\text{dec}} = \{l : \lfloor 0.70L \rfloor \leq l \leq \lfloor 0.92L \rfloor\}.$$

For graph g , condition c and permitted decision layer $l \in \mathcal{L}_{\text{dec}}$, we formed the matrix D with one row per (g, c) observation and one column per decision layer:

$$D_{(g,c),l} = dR_l^{(g,c)}, \quad l \in \mathcal{L}_{\text{dec}}.$$

The leading-trajectory support was the clean-label logit and the competing-trajectory support was the conflict-label logit; their difference defined R_l . Thus candidate labels entered the construction only as the two declared readback supports. Generated final answers, correctness labels and recorded pair-choice fields did not enter D , PCA fitting or sign orientation. After centring D , we fitted PC1 and defined

$$\Delta U_{g,c} = (D_{(g,c),:} - \bar{D})^\top v_1.$$

The sign was oriented so that the stable-condition mean exceeded the closure-condition mean; if it did not, v_1 and all scores were multiplied by -1 . ΔU is therefore a bounded dominance observable for the declared candidate pair and window, not a correctness variable. Precursor analyses excluded every column in $\mathcal{L}_{\text{dec}}$. The stricter non-R analysis also removed R_l , dR_l , candidate logits, ranks and other candidate-pair-derived predictors, retaining only predecision TopK/VIM neighbourhood-transition features. Relation-graph groups were never split across train and test folds.

Advantage-flow analyses used an observable layerwise advantage-flow proxy:

$$O_l^{\text{adv}} = dR_{l+1} - dR_l.$$

Window-level O_l^{adv} features were evaluated with relation-graph-grouped cross-validation. ΔU prediction used standardized ridge regression with $\alpha = 10$. Mechanism classification used standardized logistic regression with balanced class weights and a 1,000-iteration limit. Relation graph was the split unit.

The broad-window analysis included all transitions and model-specific initialization-to-decision windows. The predecision analysis restricted predictors to transitions before the decision window. The non-readback precursor analysis used the same decision-window ΔU target but predicted it from predecision TopK/VIM neighbourhood-transition features rather than dR_l differences.

Functional windows and the best trajectory-geometry windows were selected and summarized within the same controlled dataset. Graph-grouped folds protect prompt groups during model fitting, but they do not make this analysis-selection step independent. We therefore report complete window scans, retain checkpoint-specific multiplicity in Source Data and treat best-window values as exploratory summaries. No claim in this manuscript depends on an independently confirmed universal window.

For each scanned window, the topology objective was $1 - d_{\text{preserve}}$, where d_{preserve} is the mean within-condition preservation distance. The mechanism objective was the relation-graph-grouped mechanism-label macro-F1. The downstream objective, introduced only after ΔU was defined, was the grouped correlation

with ΔU . The archived overall scan score was

$$S_{\text{overall}} = 0.5S_{\text{topology}} + 0.8 \max(S_{\text{mechanism}}, 0) + 0.8 \max(S_{\text{downstream}}, 0).$$

Figure 4 reports topology- and mechanism-selected windows available at that stage of the discovery chain; it does not use the later downstream field to select an earlier result.

O_l^{adv} is an observable proxy for path-advantage growth. It should not be conflated with the continuous transition coordinate O_l^{cont} .

Global dominance-band control

Band-robustness control defined a global ΔU band from negative, critical and positive reference means:

$$U_- = \frac{u_{\text{neg}} + u_{\text{crit}}}{2}, \quad U_+ = \frac{u_{\text{crit}} + u_{\text{pos}}}{2}.$$

Rows were assigned to below-band, in-band or above-band phases. Figure 6d reports phase accuracy and macro-F1 from this global band midpoint classifier. This is an auxiliary band-control check, not a primary mechanism label.

One-step transition-state closure

One-step transition-closure audit audited one-step next-layer state-equation closure. The target vector was:

$$Y_{l+1} = (R_{l+1}, b_{l+1}, s_{l+1}, q_{l+1}).$$

The regression model used ridge regularization with $\alpha = 10$ and random seed 42 inside a preprocessing pipeline. Numeric predictors were median-imputed and standardized. Categorical predictors were most-frequent-imputed and one-hot encoded. All reported analyses contained at least five relation-graph groups and therefore used five-fold cross-validation grouped by relation graph. A shuffled non-grouped fallback existed for smaller diagnostic inputs, but that branch did not supply any reported score.

The main feature groups were:

Group	Features
R	Candidate margin R_l and rank gap q_l
B	Boundary distance b_l
T	Neighbourhood spread s_l
H-shape	Six local trajectory-shape terms encoding previous-step changes, velocity reversals and centre backtracking
O_l^{cont}	The first ten continuous transition-coordinate components

The reported score is macro R^2 :

$$R_{\text{macro}}^2 = \frac{1}{J} \sum_{j=1}^J R_j^2(y_j, \hat{y}_j).$$

The ten components of O_l^{cont} are latent transition coordinates derived from transition signatures. They support a local F_O state-equation description but should not be described as pure pre-transition observables.

Trajectory-geometry and residual analyses

For each observed trajectory, the layer-order geometry analysis defined the reference path as the straight chord joining its first and last analysed states, sampled at the same number of layer positions. Shuffle controls retained the same state vectors and endpoints while randomly permuting interior layer order within each trajectory; metrics were recomputed for every shuffle. This null tests whether the observed layer order contains more route organization than an order permutation. It does not match local step lengths, smoothness or autocorrelation, so it cannot distinguish depth-continuity effects from a stronger dynamical constraint. We consequently use geodesic-biased only for the reported real-versus-order-shuffle contrasts and do not claim strict equality between the real trajectory and a shortest path. Versioned implementation filenames are retained only in the code manifest.

Figure 13 display metrics are:

$$G_{\text{align}} = A_{\text{real}} - A_{\text{shuffle}}, \quad G_{\text{detour}} = \log_2 \frac{D_{\text{shuffle}}}{D_{\text{real}}}, \quad G_{\text{curvature}} = \log_2 \frac{C_{\text{shuffle}}}{C_{\text{real}}}.$$

The projection-residual analysis compares residual-only, projection-only and mixed projection-residual feature groups. Figure 14b,c reports their correlations with ΔU :

$$(\rho_{\parallel}, \rho_{\perp}, \rho_{\text{joint}}) = \text{corr}((X_{\parallel}, X_{\perp}, X_{\text{joint}}), \Delta U).$$

The parallel-perpendicular proxy analysis defines feature families for reference-aligned and structured off-direction components. Figure 14d,e reports the separate and combined proxy families:

$$X_{\text{joint}} = [X_{\parallel}, X_{\perp}].$$

Parallel and perpendicular labels are proxy component families. They should not be described as literal orthogonal decomposition in the full hidden-state manifold.

Internal-modulation replication

The internal-modulation replication used random seed 42, 96 relation graphs, batch size 4 and a 30% graph-grouped test partition. Layer windows were model-specific:

Model family	Operator layers	Precursor layers	Decision layers
Qwen	15-19	17-19	20-25
Llama	8-11	10-12	12-15
Gemma	13-17	15-17	18-23

Mechanism classifiers used the estimated signed and absolute dominance coordinates, the estimated boundary coordinate, velocity magnitude and standardized velocity, normalized layer position, a precursor-window indicator, and baseline dominance and boundary values. The random-forest classifier used 250 trees, maximum depth 7, minimum leaf size 6 and balanced-subsample class weighting; complete implementation fields and seeds are retained in the code manifest.

Mechanism labels were fixed by prompt condition before model fitting. The stable class comprised clean, redundant, irrelevant and paraphrase conditions; competition comprised weak-distractor, balanced-competition and direct-conflict conditions; closure comprised closure-update, closure-override and exception-override conditions. These labels describe the controlled intervention families and were not inferred from generated answers.

Policy classifiers used the mechanism features together with class probabilities and predicted indicators for stable, competition and closure states. The random-forest classifier used 200 trees, maximum depth 6, minimum leaf size 8 and balanced-subsample class weighting; implementation field names are retained in the code manifest.

The policy action grid was; all nonzero weights lay between 0.1 and 0.3:

$$\mathcal{A} = \{(0, 0), (0.1, 0.1), (0.2, 0.2), (0.3, 0.3), (0.3, 0), (0, 0.3), (0.3, 0.2), (0.2, 0.3)\}.$$

The fixed same-combination baseline used operator and precursor weights of 0.30.

Policy labels were fitted only on training graphs. Each training prompt was evaluated on the eight-action grid. For closure prompts, the selected label was the action with the largest signed ΔU shift. For stable and competition prompts, it was the action with the smallest absolute shift. The mechanism and policy classifiers used the same graph-level training partition; actions were then predicted for held-out graphs. No row from a held-out graph contributed to operator fitting, precursor directions, probes, dose-grid labels or either classifier. Reported classifier accuracy and macro-F1 are training diagnostics and are not presented as held-out generalization estimates.

Specificity was computed as:

$$S = \Delta U - \Delta U_{\text{base}}, \quad \text{specificity} = E(|s| \mid M = \text{closure}) - E(|s| \mid M \neq \text{closure}).$$

The 50-shuffle replication used three matched null families: mechanism-policy pairing shuffles, policy-label shuffles and joint shuffles. Mechanism-label shuffles used seeds beginning at 1,042, and policy-label shuffles used seeds beginning at 2,042; each replicate incremented the corresponding seed by one. Joint shuffles used a disjoint seed range before applying the same offsets. Exact seed construction is retained in the code manifest.

In the manuscript, this experiment is described as internal trajectory-dominance modulation. The endpoint does not measure task correctness or deployment control.

Independent task-family controls

The lexical-category and single-step addition tasks each comprised 64 problems under six matched prompt conditions. Complete problem groups were split into 44 training and 20 held-out groups. A task-specific candidate-separation coordinate was fitted on training problems only and evaluated against 200 training-label shuffles; the numerical relation-graph ΔU axis was not reused.

The non-binary benchmark control used a deterministic 96-item ARC-Challenge validation subset. It comprised the earliest dataset-order rows with exact A–D answer keys, a rule fixed without reference to model outputs to bound extraction cost and prevent outcome-based selection. Every question and all four answer choices were presented through the native chat template. Lexical, addition and ARC geometry used normalized last-token hidden states and 50 endpoint-preserving order shuffles. No model-specific window search was conducted for these geometry tests.

Prospective final candidate-margin forecast

The frozen relation-graph split contained 67 training groups (670 prompts) and 29 held-out groups (290 prompts). At each decision-window cutoff, ridge regression with penalty 10 forecast the exact final clean-minus-conflict output-logit margin from the current and earlier margins only. A balanced logistic model with $C = 1$ predicted its sign from the same features. No later state, realized transition,

generated token or correctness label entered the predictors. The order null retained the current-cutoff value and independently permuted all earlier values within each prompt. Fifty null datasets were refitted and evaluated on the unchanged held-out graphs. Confidence intervals used 1,000 graph-bootstrap replicates.

Cross-architecture random-initialization control

For each Qwen2.5-1.5B, Llama-3.2-1B and Gemma-2-2B configuration, three models were instantiated without trained tensors using seeds 1701, 1702 and 1703. Native tokenizers, the same 96 relation prompts, extraction rule, normalization and 50 endpoint-preserving shuffles were held fixed. The matched initialization analysis used null seed 20260717. The control tests whether chord-directed order requires trained parameters in these configurations; it does not show functional equivalence or identify a universal architecture–training decomposition.

Out-of-fold displacement-to-boundary bridge

For prompt p , the margin and trajectory displacements were $\delta M_p = M_{p,\text{policy}} - M_{p,0}$ and $\delta \Delta U_p = \Delta U_{p,\text{policy}} - \Delta U_{p,0}$. Checkpoint- and regime-specific affine ridge models with penalty 10^{-6} estimated $\widehat{\delta M}_p = a_{m,r} + b_{m,r} \delta \Delta U_p$. Five-fold grouped cross-validation held out complete relation graphs, and an inner graph cross-fit supplied training predictions for a one-variable logistic calibrator. Continuous and crossing intervals used 1,000 graph-bootstrap replicates with seed 20260714. Correctness labels did not enter either model.

Confidence-gated emitted-token boundary control

The output-boundary analysis was run independently in Qwen, Llama and Gemma using the same seed-42 graph split: 67 training graphs and 29 held-out graphs, containing 87 closure and 203 non-closure held-out prompts. Intervention layers were 15–19, 8–11 and 13–17, respectively. Separation directions, transition operators, precursor directions, mechanism classifiers, action targets and boundary policies were fitted on training graphs only. Generated answer text, correctness and held-out outcomes entered neither fitting nor gate construction.

The bounded action library was

$$\mathcal{A} = \{(0, 0), (0.3, 0.3), (0.6, 0.3), (0.6, 0.6), (0.9, 0.3), (0.9, 0.6), (0.9, 0.9), (1.0, 0.6), (1.0, 0.9), (1.0, 1.2)\}.$$

For closure prompts, training utility was candidate-margin displacement plus a 5-unit crossing bonus minus $0.03(\alpha^2 + \beta^2)$; for non-closure prompts it was negative absolute margin displacement minus $0.10(\alpha^2 + \beta^2)$. The supervised policy used the same 15 pre-intervention features as the internal-control analysis.

For intervention layers L_I , mean closure posterior and normalized posterior entropy were $\bar{\pi}_p = |L_I|^{-1} \sum_{l \in L_I} \pi_{p,l}(\text{closure})$ and $\bar{H}_p = |L_I|^{-1} \sum_{l \in L_I} [-\sum_k \pi_{p,l}(k) \log \pi_{p,l}(k)] / \log 3$. The predeclared gate was $g_p = I[\bar{\pi}_p \geq 0.50 \wedge \bar{H}_p \leq 0.90]$. Closed gates applied (0, 0) at every intervention layer. The primary endpoint was a strict full-vocabulary top-1 continuation change from the declared conflict token to the declared clean token. Non-closure top-1 changes and state-norm ratios were damage guards.

Each of 50 safety-matched repeats permuted the complete model-specific action vector without fixed points among gate-open prompts within the same closure condition. The null preserved the gate-open set, gate-closed zero actions and every condition-by-layer action multiset. One-sided empirical values were $(1 + n_{\text{null} \geq \text{real}}) / (1 + 50)$. This null tests trajectory-to-dose assignment, not whether the gated actuator family can cross the boundary.

Falsification-first scaffold and execution controls

The relational-continuity control propagated prompt states through random fixed maps matched to each architecture’s dimensional schedule. Qwen, Llama and Gemma configurations were crossed with lexical, addition and four-choice ARC prompts and two seeds, yielding 18 declared conditions. A condition passed when the prespecified real-order continuity contrast was positive relative to its matched intermediate-order null. This tests architecture-compatible propagation, not task semantics or trained function.

The execution-order audit used the same Qwen prompts and weights under native, reversed and fixed-permutation block orders. Geometry, full-vocabulary top-1 agreement, declared-candidate agreement and Jensen–Shannon divergence were computed against native execution. The metric audit used Llama trajectories decomposed into normalized direction and state norm. Direction-only, norm-only fixed-direction and frozen radial-exponent variants were evaluated under seven prespecified task gates. These controls distinguish finite-path metric regularity from functional execution and radial sensitivity.

Incremental path-history tests

For each task-checkpoint condition, the current-state model was compared with a model augmented by ordered-prefix summaries under group-held-out evaluation and matched order nulls. The initial audit contained 18 lexical and addition conditions. A separate mixed-arithmetic replication used nine independently generated conditions. Passing required a positive held-out incremental gain above the prespecified null threshold. No condition was redefined after inspecting outcomes.

Pythia component-compatibility swaps

Pythia-160m checkpoints at initialization and after training supplied embeddings (E), transformer blocks (B), final normalization (N) and output head (W). All $2^4 = 16$ initialized/trained component combinations were assembled and evaluated on the same 53 held-out lexical cloze prompts. Main effects and higher-order interactions were estimated from candidate margins across the complete factorial design. The superadditive margin excess compared the fully trained system with the additive prediction from component-wise gains; its interval used grouped bootstrap resampling over prompts.

Training-time coordinate audit

Seven Pythia checkpoints were evaluated with two complementary coordinates. The model-native candidate coordinate used each checkpoint’s own output representation and tested order gain against matched shuffles. The fixed-coordinate analysis fitted one linear coordinate at the final checkpoint and applied it without refitting to earlier checkpoints. Macro F1 was evaluated on held-out lexical prompts. A coordinate refitted within each checkpoint served as a negative control: because it already separated categories at initialization, it was not accepted as evidence that training created the coordinate. These tests identify when task-aligned coordinate compatibility becomes available; they do not reconstruct a complete training-shaped representational geometry.

Software environments

Python GPU and model analyses used a frozen Conda environment containing Python 3.11.15, PyTorch 2.5.1+cu121, CUDA 12.1 and Transformers 5.9.0. NumPy 2.4.4, pandas 3.0.3, scikit-learn 1.8.0 and SciPy 1.17.1 supplied the numerical and statistical routines. CUDA was available on one NVIDIA GeForce RTX 4080 SUPER GPU.

Figures 1, 2 and 4-15 and Supplementary Figs. S1-S12 were rendered in a frozen Conda environment with R 4.3.3, ggplot2 3.5.2, ggforce 0.5.0, ggnewscale 0.5.2, patchwork 1.3.2, ggrepel 0.9.6 and cowplot 1.2.0. Data handling used dplyr 1.1.4, tidyr 1.3.1 and readr 2.1.5; SVG and raster export used svglite 2.2.1 and ragg 1.5.0. Figure 3 v0.2 was regenerated after restoration of its source script with R 4.6.1 and ggplot2 4.0.3; its session information is retained beside the figure and in the Source Data manifest. Both environments passed their recorded verification scripts.

Repository and code status

The public release package contains processed Source Data, machine-readable metadata, analysis scripts, figure scripts and environment manifests. Original analysis scripts are retained for provenance, with repository-relative wrappers where workstation-specific paths were present. The Source Data and code archives are identified by their Zenodo DOIs in the Data and Code availability statements and are mirrored in the linked public GitHub reproducibility branch. File-level correspondence is documented in the Source Data and code manifests rather than in the manuscript text.

Claim boundaries

The interpretation follows seven constraints. ΔU is a bounded dominance observable. Vocabulary readbacks are diagnostics. O_l^{adv} and O_l^{cont} denote different quantities. Low-dimensional plots are orientation aids. The trajectory-geometry analysis supports chord-directed order relative to declared nulls, not an exact geodesic, learned reasoning signature or universal law. Internal modulation and reachability of the declared emitted-token candidate boundary are established across the three tested checkpoints; fine-grained action assignment, answer correctness, open-ended generation control and deployment safety are not established.

AI-assisted manuscript preparation

AI-assisted tools were used to organize notes, debug code, check terminology and claim boundaries, and prepare R plotting scripts from author-defined specifications. H.Z. set the research questions, designed and ran the experiments, checked numerical results and figures against Source Data, interpreted the findings and decided which claims to retain or withdraw. Y.W., R.S. and W.C. contributed the domain and methodological support described in the Author contributions statement, and all authors reviewed the manuscript. The authors take responsibility for the manuscript. No scientific conclusion was accepted solely from AI output, and no publication figure was produced by an image-generation system.

9 Data availability

Processed panel-level tables are archived under the Zenodo source-data concept DOI <https://doi.org/10.5281/zenodo.21317355> and mirrored in the public project repository at https://github.com/zhaozheng1990-debug/Cognitive_Theory_And_LLM_PhaseMap. The v0.53 release adds row-level evidence for random fixed-map continuity, reordered execution, incremental path-history tests, radial metric sensitivity, Pythia component swaps and training-time coordinate emergence. It retains the v0.52 task-family, forecasting and cross-checkpoint output-boundary controls. The release contains prompt and checkpoint metadata, random seeds, null specifications, figure contracts and file-level checksums. Raw hidden-state arrays are large derivatives of third-party checkpoints and are not redistributed; prompts, revisions and extraction

scripts required to regenerate them are provided subject to checkpoint access and licensing constraints. Model weights are not redistributed.

10 Code availability

Analysis scripts, deterministic R figure scripts, environment records, smoke tests and portable path templates are archived under the Zenodo code concept DOI <https://doi.org/10.5281/zenodo.21317440> and mirrored in the public reproducibility branch. The v0.53 code release adds falsification-first scaffold, execution-order, path-history, radial-metric and Pythia training workflows to the v0.52 release. Workstation-specific paths are replaced or wrapped with repository-relative arguments while provenance-preserving source files are retained. Full experiment reruns require the listed model checkpoints and upstream datasets that cannot be redistributed.

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13 Author contributions

H.Z. conceived the research, defined the questions, formulated the hypotheses developed during the investigation, designed and ran the experiments, established the evidential tests and revised interpretations in response to the results. Y.W. contributed artificial-intelligence domain knowledge and methodological guidance. R.S. contributed operations-research and mathematical-method support. W.C. contributed discussion, advice and conceptual inspiration from medical, neuroscience and cognitive-science perspectives.

Y.W., R.S. and W.C. participated in their personal capacities as independent research collaborators and did not provide employer data or resources. All authors reviewed and approved the manuscript. AI-assisted operations are disclosed in Methods and do not constitute authorship.

14 Competing interests

The authors declare no competing interests.

15 Additional information

Supplementary information accompanies this manuscript. Correspondence and requests for materials should be addressed to Heng Zhao (zhaoheng1990@gmail.com).

A Discovery and execution chronology

Hidden-state trajectories emerged late in the study. The starting difficulty was practical: intermediate activations were available, with no obvious way to inspect their change across layers. TopK vocabulary readbacks supplied a common readout at each layer and gave the first stable observational surface.

That surface soon proved more useful for asking questions than for settling them. Adjacent readbacks could be fitted by measurable transition maps, and their signatures appeared to cluster into a small number of regimes. The resulting boundary-bulk-boundary picture was appealing but too literal. Fine cluster labels tracked depth strongly, whereas only a coarser late transition survived the layer-clock controls. Different checkpoints also placed comparable structure at different depths, which made model-specific functional windows more defensible than a common layer index.

The next difficulty concerned interpretation. A low-rank coordinate, ΔU , summarized candidate-relative variation within those windows. Predecision controls found related structure earlier in the stack. The source of the readback geometry was still unclear. Learned vocabulary rows seemed a plausible explanation, yet row permutation left preservation largely intact. Matched Gaussian projections often preserved as much or more. The readback remained useful after its learned-row explanation failed.

This failure prompted the direct comparison that changed the object of the paper. Hidden-state and trajectory-shape variables carried more information about later organization than the vocabulary-facing alternatives in the tested grouped analyses. We therefore followed the ordered hidden states themselves. Transition coordinates, ΔU , geometric summaries and residual features are measurements of this trajectory, not replacements for it.

The choice remains provisional in the scientific sense. A later strict-predecision matched-capacity audit retained the hidden-state advantage for controlled mechanism labels but found that ΔU prediction depended on checkpoint and retained dimension. High-dimensional layer-state trajectories therefore remain the organizing empirical object, without a claim that they outperform readbacks on every metric. This qualified conclusion emerged because an earlier explanation failed.

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